



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Scienze Fisiche,
Informatiche e Matematiche

11. Ottimizzazioni sulla memoria

Linguaggi e Compilatori [I215-011]

Corso di Laurea in INFORMATICA
(D.M.270/04) [16-262]
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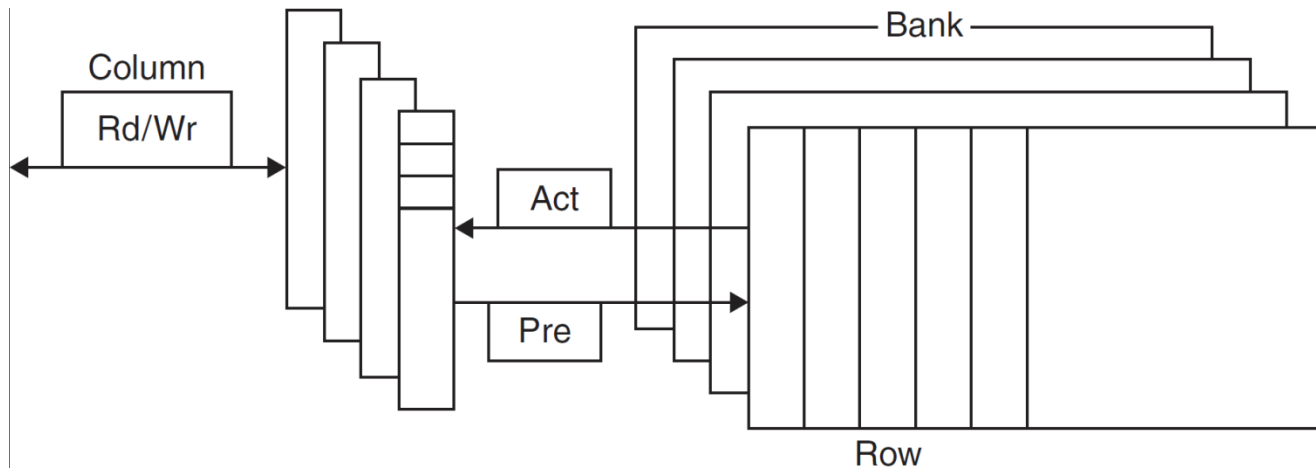
Credits

- Hennessy, Patterson, “Computer Organization and Design - RISC-V Edition : The Hardware Software Interface”, Morgan Kaufmann
- Marchetti-Spaccamela, La Sapienza Università di Roma, “Cache e gerarchia di memoria”
- Gibbons, Carnegie Mellon University, “Optimizing Compilers”
- Pekhimenko, University of Toronto, “Compiler Optimization”
- Juurlink, Technische Universität Berlin, “Advanced Memory Hierarchy Design”

Outline

DRAM Technology

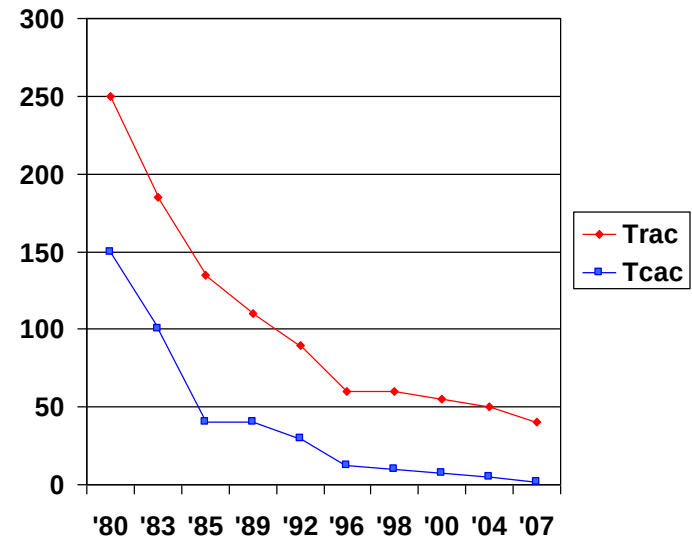
- Data stored as a charge in a capacitor
 - Single transistor used to access the charge
 - Must periodically be refreshed
 - Read contents and write back
 - Performed on a DRAM "row"



Advanced DRAM Organization

- Bits in a DRAM are organized as a rectangular array
 - DRAM accesses an entire row
 - Burst mode: supply successive words from a row with reduced latency
- Double data rate (DDR) DRAM
 - Transfer on rising and falling clock edges
- Quad data rate (QDR) DRAM
 - Separate DDR inputs and outputs

DRAM Generations

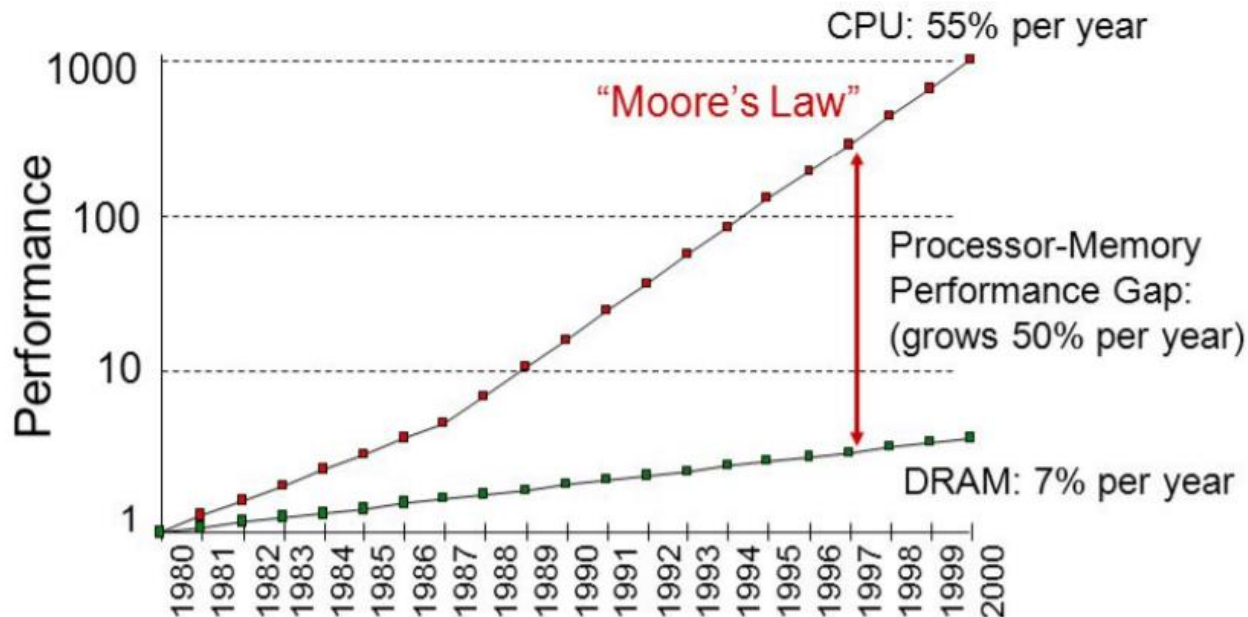


Year introduced	Chip size	\$ per GiB	Total access time to a new row/column	Average column access time to existing row
1980	64 Kibibit	\$1,500,000	250 ns	150 ns
1983	256 Kibibit	\$500,000	185 ns	100 ns
1985	1 Mebibit	\$200,000	135 ns	40 ns
1989	4 Mebibit	\$50,000	110 ns	40 ns
1992	16 Mebibit	\$15,000	90 ns	30 ns
1996	64 Mebibit	\$10,000	60 ns	12 ns
1998	128 Mebibit	\$4,000	60 ns	10 ns
2000	256 Mebibit	\$1,000	55 ns	7 ns
2004	512 Mebibit	\$250	50 ns	5 ns
2007	1 Gibibit	\$50	45 ns	1.25 ns
2010	2 Gibibit	\$30	40 ns	1 ns
2012	4 Gibibit	\$1	35 ns	0.8 ns

DRAM Performance Factors

- Row buffer
 - Allows several words to be read and refreshed in parallel
- Synchronous DRAM
 - Allows for consecutive accesses in bursts without needing to send each address
 - Improves bandwidth
- DRAM banking
 - Allows simultaneous access to multiple DRAMs
 - Improves bandwidth

The memory wall



CPU
60% per yr
2X in 1.5 yrs

DRAM
9% per yr
2X in 10 yrs

- ❖ 1980 – No cache in microprocessor
- ❖ 1995 – Two-level cache on microprocessor

- access to data is a limiting factor
 - *we'd better spend transistors to improve memory behavior*

Are there other "types" of memory?

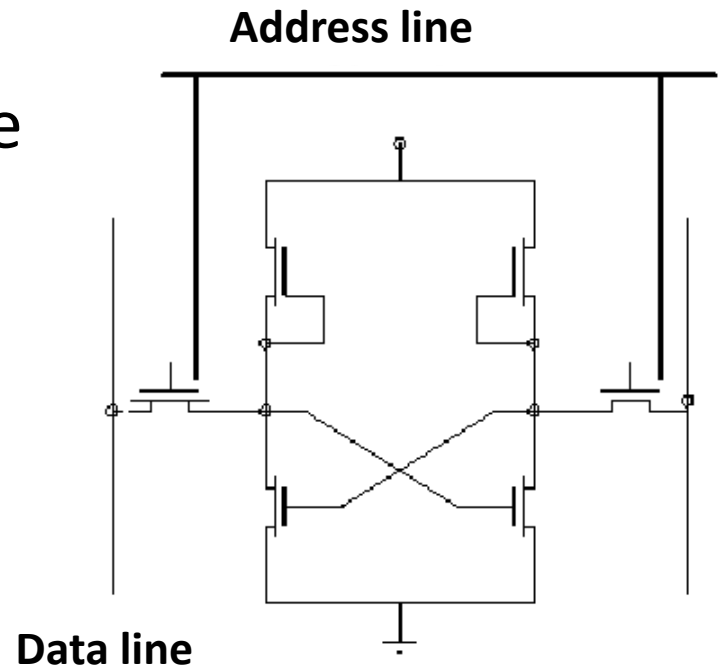
- Different technology than DRAM?
- With better performance than DRAM?

Static RAM (SRAM)

The elementary cell is composed of 6 MOS transistors forming a FLIP-FLOP

The information remains stable as long as the circuit is powered

- Fast access time
- High density
- High cost



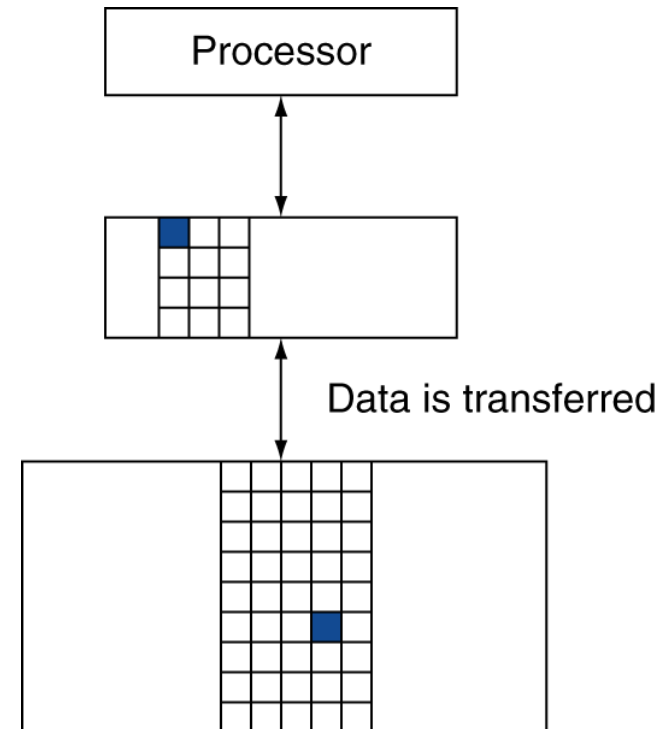
Memory Technology

Memory technology	Typical access time	\$ per GiB in 2012
SRAM semiconductor memory	0.5–2.5 ns	\$500–\$1000
DRAM semiconductor memory	50–70 ns	\$10–\$20
Flash semiconductor memory	5,000–50,000 ns	\$0.75–\$1.00
Magnetic disk	5,000,000–20,000,000 ns	\$0.05–\$0.10

- Ideal memory
 - Access time of SRAM
 - Capacity and cost/GB of disk

Memory Hierarchy Levels

- Block (aka line): unit of copying
 - May be multiple words
- If accessed data is present in upper level
 - Hit: access satisfied by upper level
 - Hit ratio: hits/accesses
- If accessed data is absent
 - Miss: block copied from lower level
 - Time taken: miss penalty
 - Miss ratio: misses/accesses
 $= 1 - \text{hit ratio}$
 - Then accessed data supplied from upper level



Principle of Locality

- Programs access a small proportion of their address space at any time
- Temporal locality
 - Items accessed recently are likely to be accessed again soon
 - e.g., instructions in a loop, induction variables
- Spatial locality
 - Items near those accessed recently are likely to be accessed soon
 - E.g., sequential instruction access, array data

Locality in programs



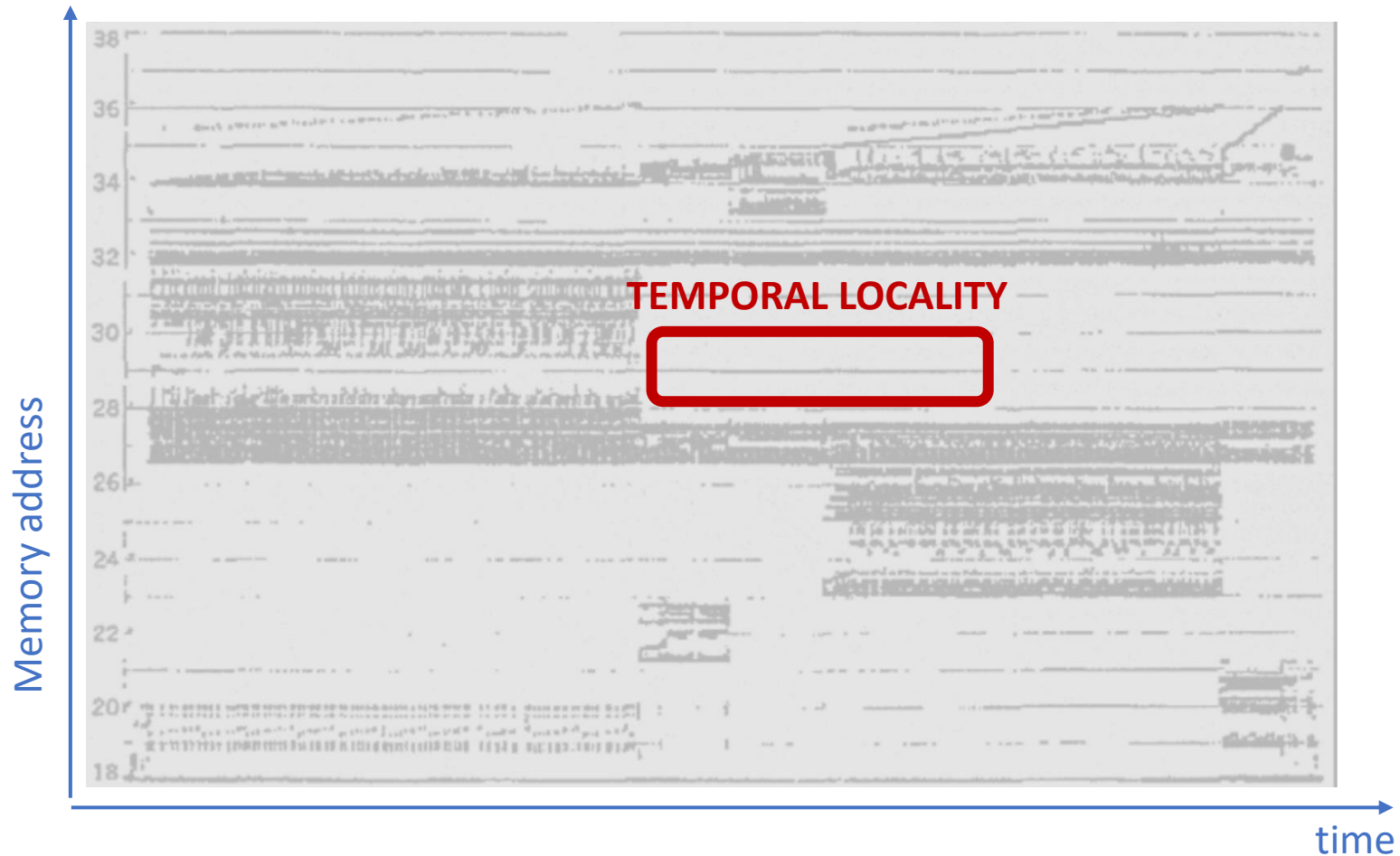
Donald J. Hatfield, Jeanette Gerald: Program Restructuring for Virtual Memory. IBM Systems Journal 10(3): 168-192 (1971)

Locality in programs



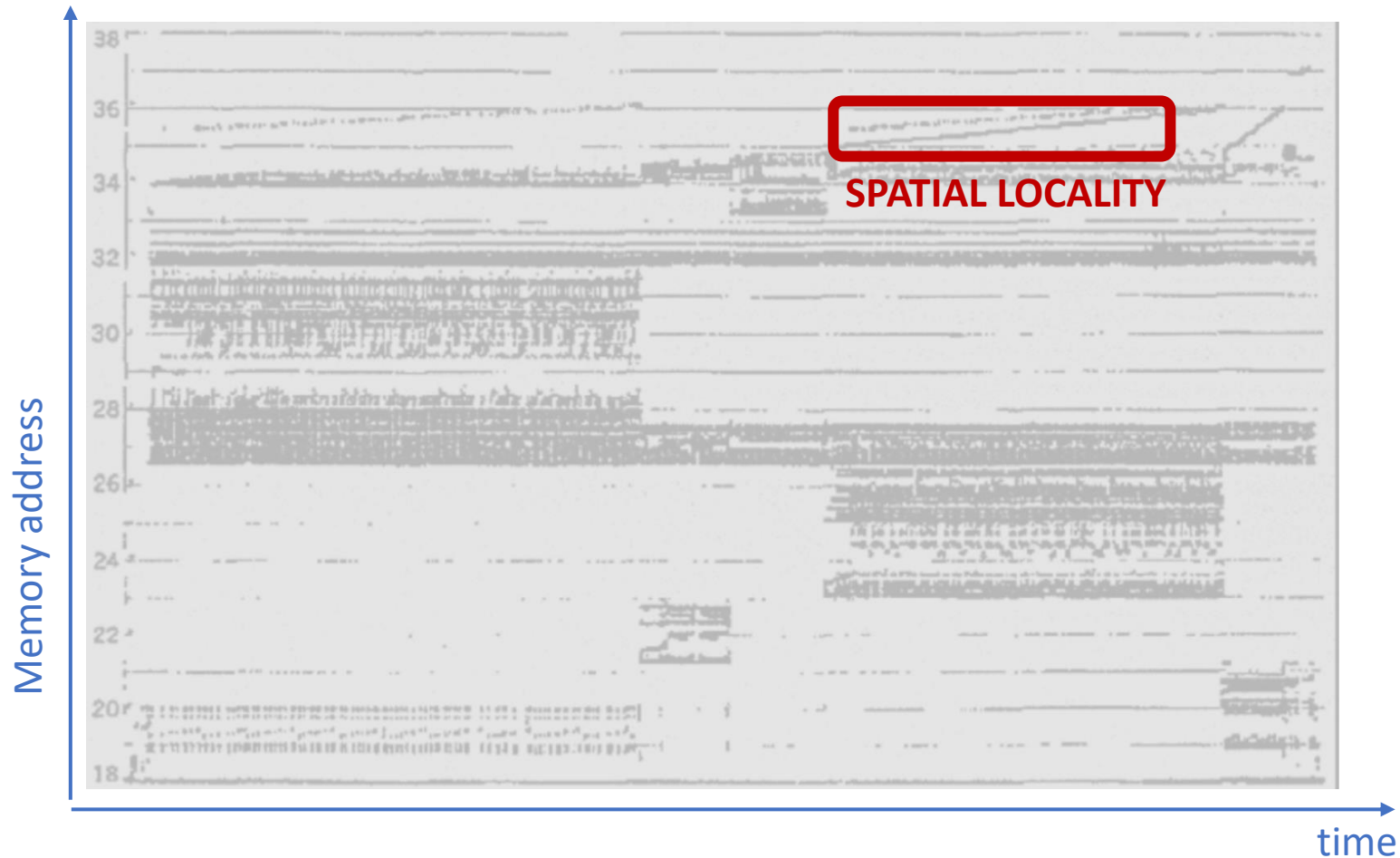
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Taking Advantage of Locality

- Memory hierarchy
- Store everything on disk
- Copy recently accessed (and nearby) items from disk to smaller DRAM memory
 - Main memory
- Copy more recently accessed (and nearby) items from DRAM to smaller SRAM memory
 - Cache memory attached to CPU

Caches

- The term "cache" was used for the first time to indicate the level of the memory hierarchy between the CPU and the main memory
 - but it is nowadays used to indicate any memory management scheme that exploits access locality
- Simple mechanism: The processor only interacts with the cache. Upon cache miss the requested word is copied from the main memory to the cache
- Need mechanisms for:
 - Knowing if data is present in the cache
 - If so, locate the data and access it
- These operations must execute as fast as possible, as this heavily impacts the performance of the memory system

Hit & Miss

- We have a **cache hit** when the address requested from the level above (e.g., the processor) are present in the cache.
- We have a **cache miss** when the address requested from the level above is not present in the cache and must be fetched from the level below.
- A key performance metric for the memory system is the **hit rate**, i.e., the percentage of memory accesses that hit in the cache.
- The **miss rate** is defined as *1-hit rate* and indicates the percentage of memory accesses that miss in the cache.
- **Hit time**: time required to fetch the data from the cache in case of a hit. Includes the time needed to determine if the access is a hit or a miss.
- **Miss time**: time required to replace data at the requested address from the lower level of the hierarchy.

Cache Misses

- On cache hit, CPU proceeds normally
- On cache miss
 - Stall the CPU pipeline
 - Fetch block from next level of hierarchy
 - Instruction cache miss
 - Restart instruction fetch
 - Data cache miss
 - Complete data access

Cache Misses

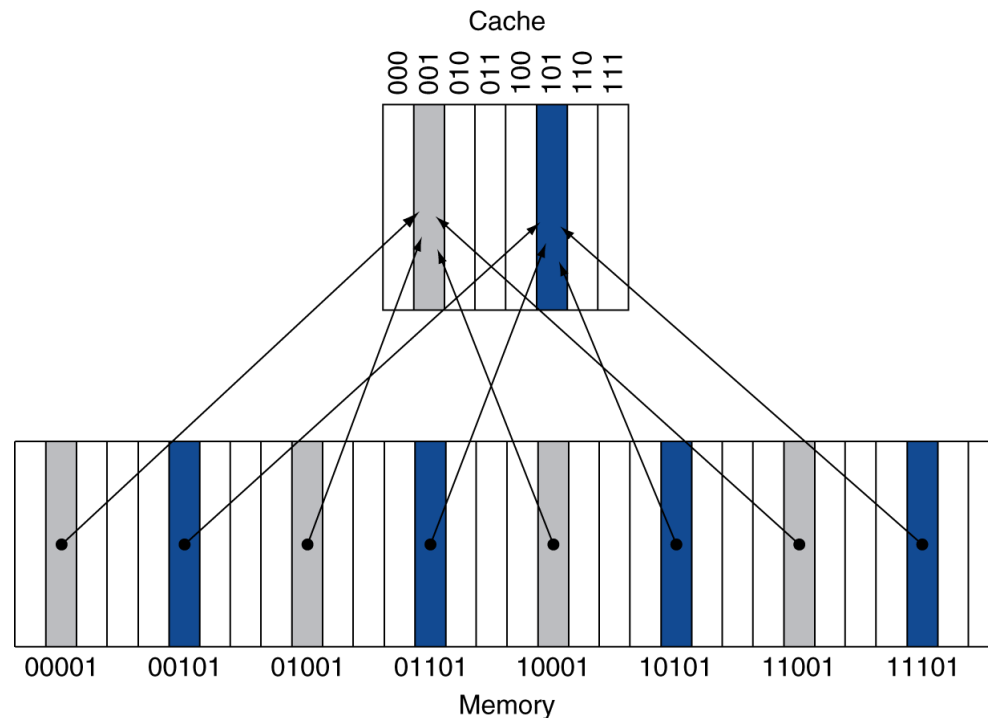
- Three types of cache miss in single-core processors:
- Compulsory (cold cache)
 - Happens when no data is present in the cache (typically at program startup)
- Conflict
 - Happens when two addresses in DRAM are mapped on the same cache line
- Capacity
 - Happens when there is no space left in the cache

Types of caches

- Direct-mapped caches
- Associative caches
- Which pros and cons?

Direct Mapped Cache

- Location determined by address
- Direct mapped: only one choice
 - (Block address) modulo (#Blocks in cache)



- #Blocks is a power of 2
- Use low-order address bits

Tags and Valid Bits

- How do we know which particular block is stored in a cache location?
 - Store block address as well as the data
 - Actually, only need the high-order bits
 - Called the tag
- What if there is no data in a location?
 - Valid bit: 1 = present, 0 = not present
 - Initially 0

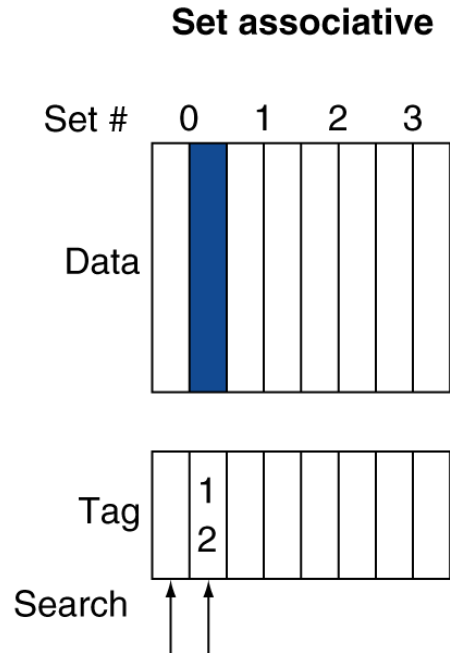
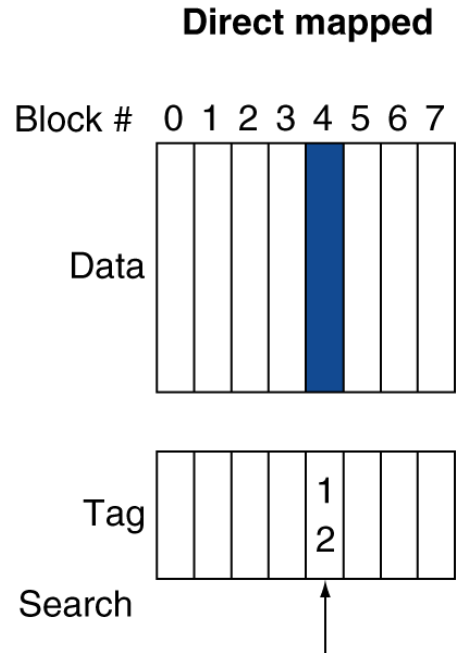
Direct Mapped Cache

- PROs
 - Very simple addressing scheme
 - Very fast hit time
- CONs
 - Subject to conflict misses
 - Non-optimal use of cache space

Associative Caches

- Fully associative
 - Allow a given block to go in any cache entry
 - Requires all entries to be searched at once
 - Comparator per entry (expensive)
- n -way set associative
 - Each set contains n entries
 - Block number determines which set
 - (Block number) modulo (#Sets in cache)
 - Search all entries in a given set at once
 - n comparators (less expensive)

Associative Cache Example



Spectrum of Associativity

- For a cache with 8 entries

One-way set associative (direct mapped)

Block	Tag	Data
0		
1		
2		
3		
4		
5		
6		
7		

Two-way set associative

Set	Tag	Data	Tag	Data
0				
1				
2				
3				

Four-way set associative

Set	Tag	Data	Tag	Data	Tag	Data	Tag	Data
0								
1								

Eight-way set associative (fully associative)

[illegible]

Associative Caches

- PROs
 - Better use of cache space
 - Less prone to conflict miss
- CONs
 - More expensive addressing scheme

Exploiting spatial locality

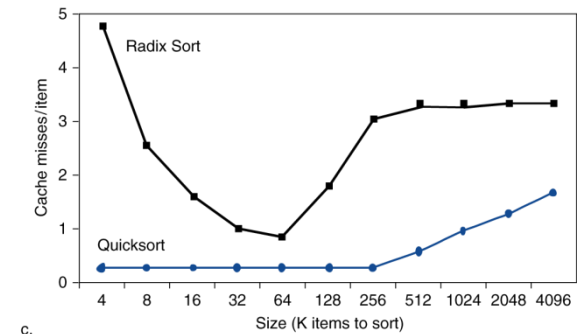
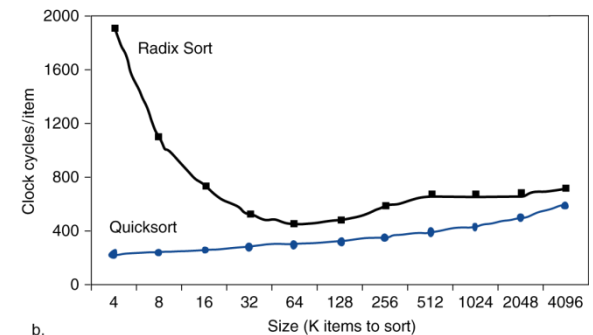
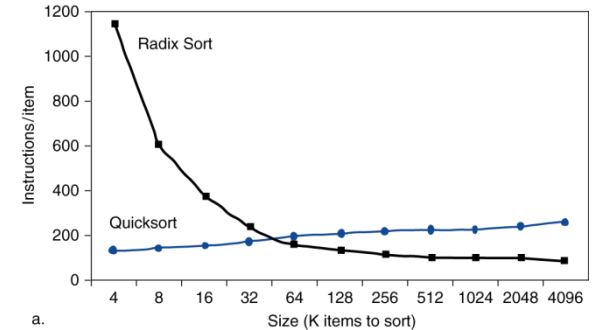
- To exploit spatial locality cache blocks (or lines) must be larger than a single word.
 - Upon a *miss* several adjacent words will be loaded in the cache line, which have a high probability of being requested in the future.

Block Size Considerations

- Larger blocks should reduce miss rate
 - Due to spatial locality
- But in a fixed-sized cache
 - Larger blocks $\Rightarrow$ fewer of them
 - More competition $\Rightarrow$ increased miss rate
 - Larger blocks $\Rightarrow$ pollution
- Larger miss penalty
 - Can override benefit of reduced miss rate
 - Early restart and critical-word-first can help

Interactions with Software

- Misses depend on memory access patterns
 - Algorithm behavior
 - Compiler optimization for memory access



Cache summary

- **Cache hit:**
 - in-cache memory access—cheap
- **Cache miss:**
 - non-cached memory access—expensive
 - need to access next, slower level of hierarchy
- **Cache line size:**
 - # of bytes loaded together in one entry
 - typically a few machine words per entry
- **Capacity:**
 - amount of data that can be simultaneously in cache
- **Associativity:**
 - direct-mapped: only 1 address (line) in a given range in cache
 - n-way: $n \geq 2$ lines w/ different addresses can be stored



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Compiler optimizations for the Memory Hierarchy

Memory Hierarchy Optimizations

- Reduce *memory latency*
 - The latency of a memory access is the time (usually in cycles) between a memory request and its completion
- Maximize *memory bandwidth*
 - Bandwidth is the amount of useful data that can be retrieved over a time interval
- Manage overhead
 - Cost of performing optimization (e.g., copying) should be less than anticipated gain

Reuse and Locality

- Consider how data is accessed
 - **Data reuse:**
 - Same or nearby data used multiple times
 - Intrinsic in computation
 - **Data locality:**
 - Data is reused and is present in “fast memory”
 - Same data or same data transfer
- If a computation has reuse, what can we do to get locality?
 - Appropriate data placement and layout
 - Code reordering transformations

Optimizing Cache Performance

- Things to enhance:
 - temporal locality
 - spatial locality
- Things to minimize:
 - conflicts (i.e. bad replacement decisions)

What can the *compiler* do to help?

Two Things We Can Manipulate

- Time:
 - When is an object accessed?
- Space:
 - Where does an object exist in the address space?

How do we exploit these two levers?

Time: Reordering Computation

- What makes it difficult to know *when* an object is accessed?
- How can we predict a *better time* to access it?
 - What information is needed?
- How do we know that this would be *safe*?

Space: Changing Data Layout

- What do we know about an object's **location**?
 - scalars, structures, pointer-based data structures, arrays, code, etc.
- How can we tell what a **better layout** would be?
 - how many can we create?
- To what extent can we **safely** alter the layout?

Types of Objects to Consider

- Scalars
- Structures & Pointers
- Arrays

Scalars

- Locals
- Globals
- Procedure arguments
- Is cache performance a concern here?
- If so, what can be done?

```
int x;  
double y;  
foo(int a) {  
    int i;  
  
    ...  
    x = a*i;  
    ...  
}
```

Structures and Pointers

- What can we do here?
 - within a node
 - across nodes

```
struct {  
    int count;  
    double velocity;  
    double inertia;  
    struct node *neighbors[N];  
} node;
```

- What limits the compiler's ability to optimize here?

Arrays

```
double A[N][N], B[N][N];
```

```
...
```

```
for i = 0 to N-1
```

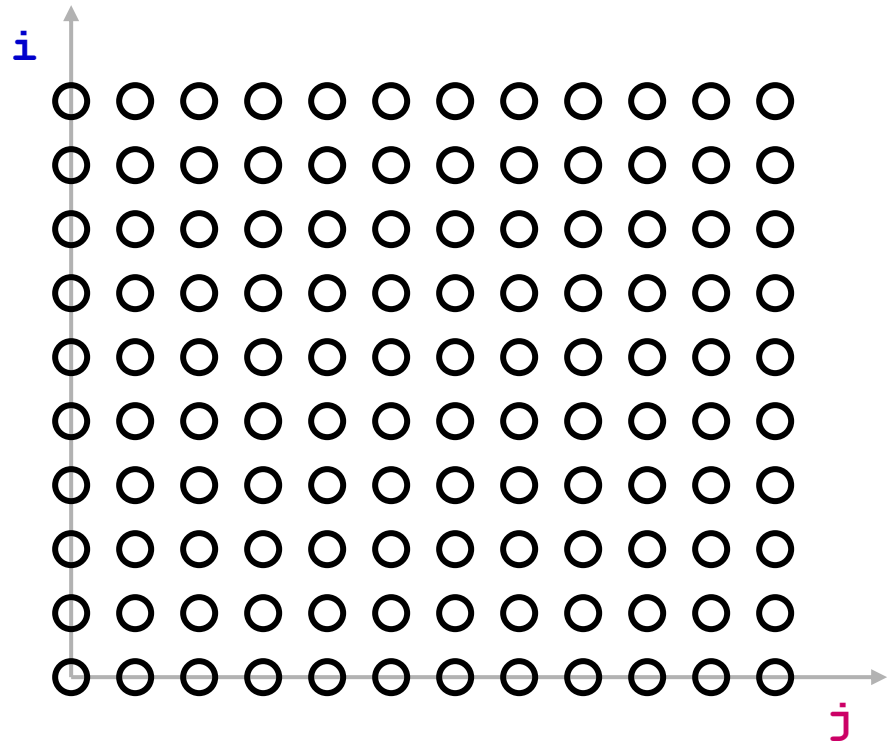
```
    for j = 0 to N-1
```

```
        A[i][j] = B[j][i];
```

- usually accessed within **loops nests**
 - makes it easy to understand “time”
- what we know about **array element addresses**:
 - start of array?
 - relative position within array

Iteration Space

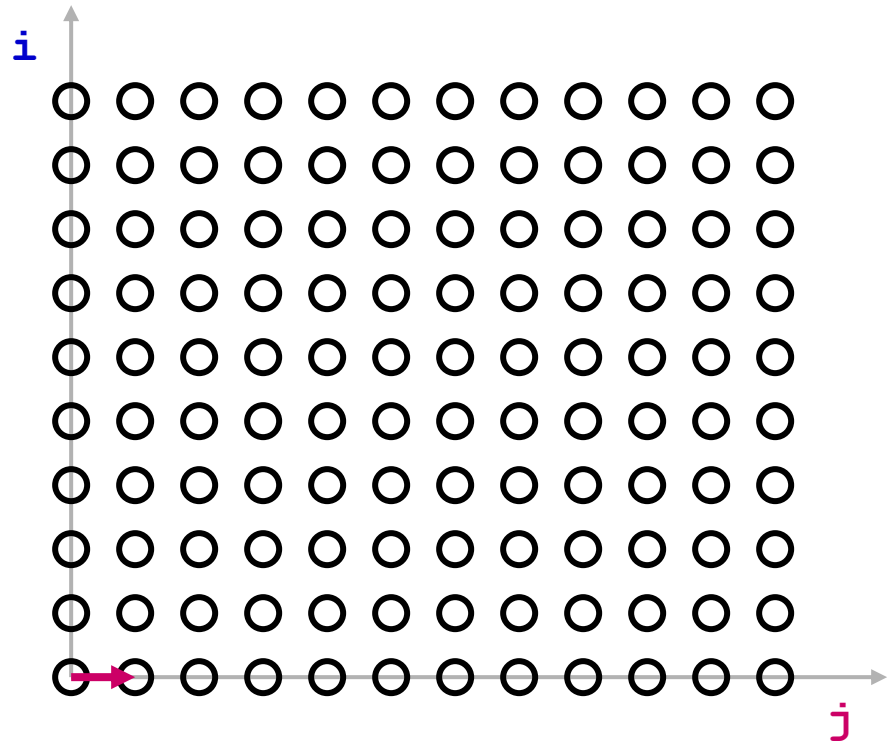
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- each position represents an iteration

Visitation Order in Iteration Space

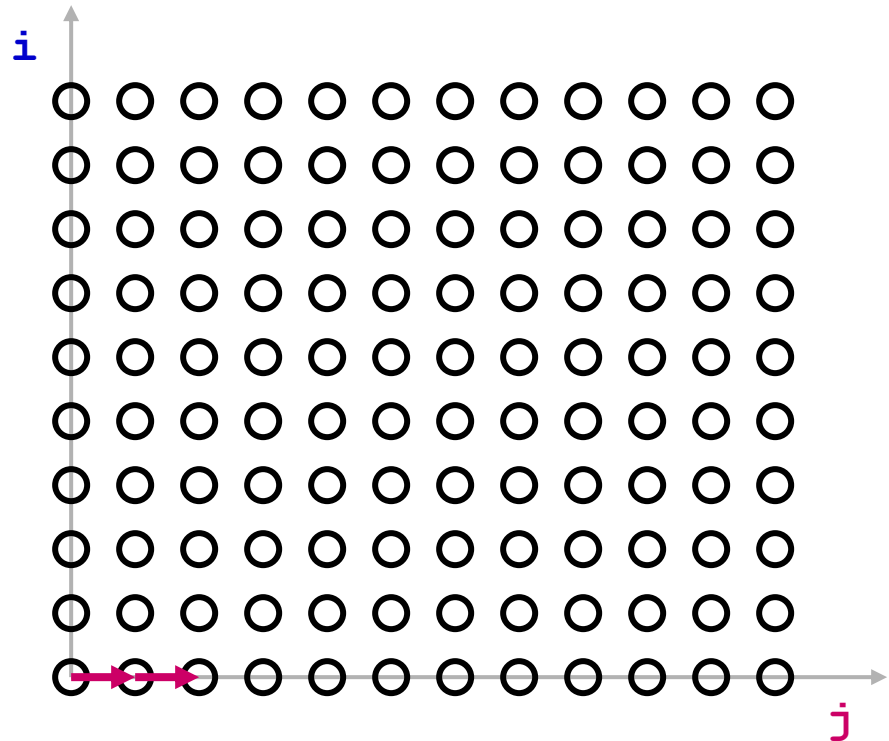
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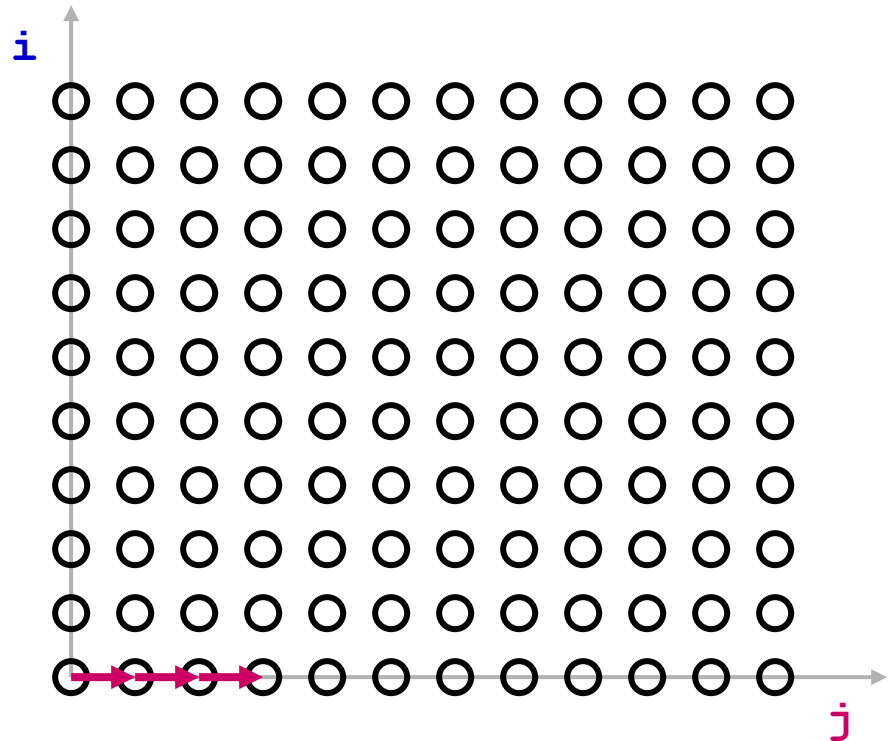
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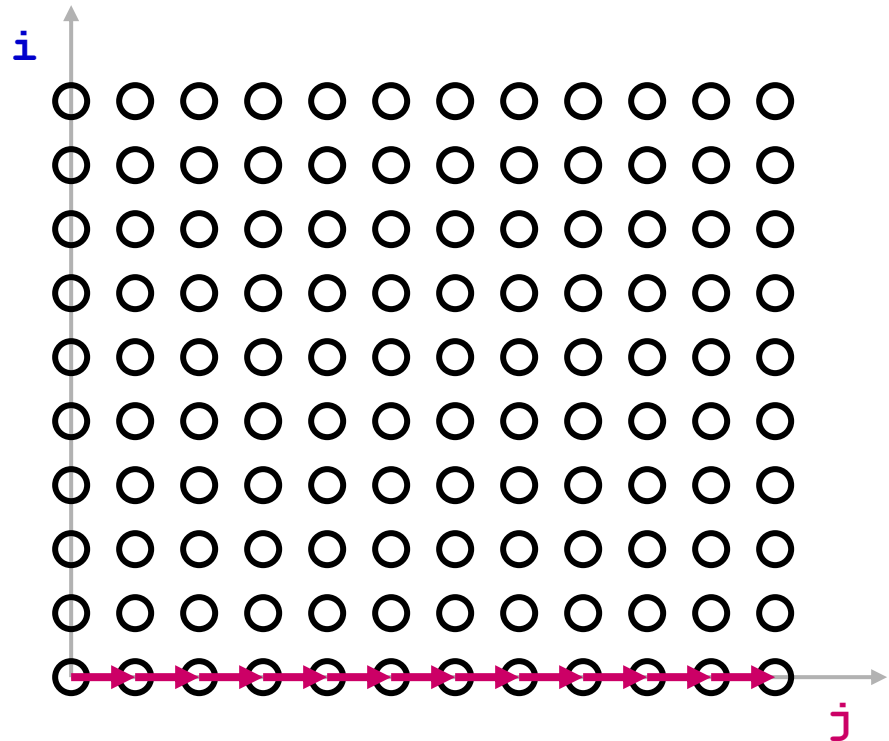
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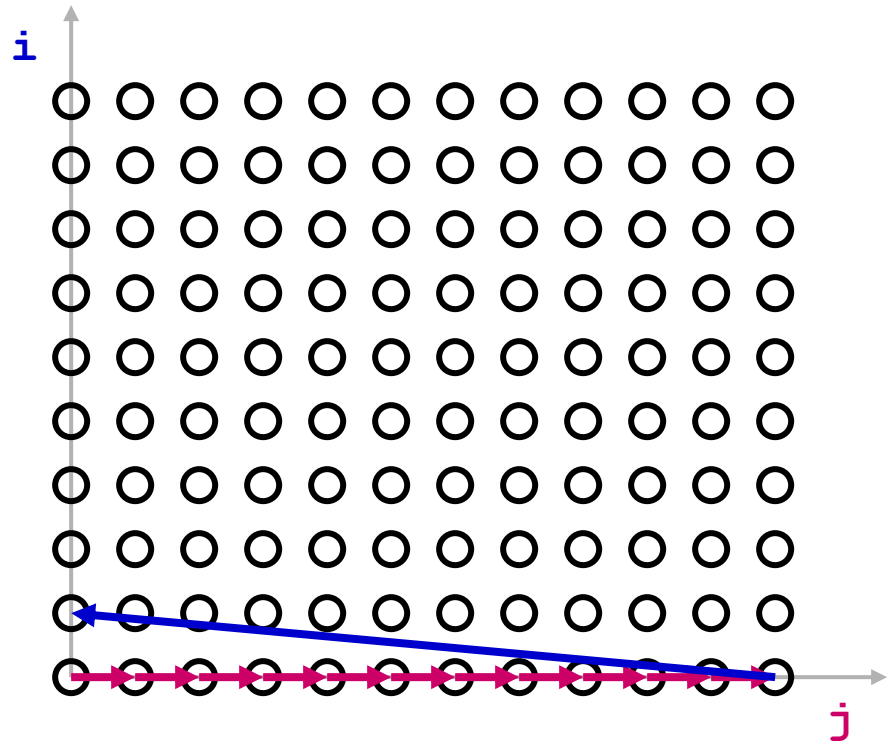
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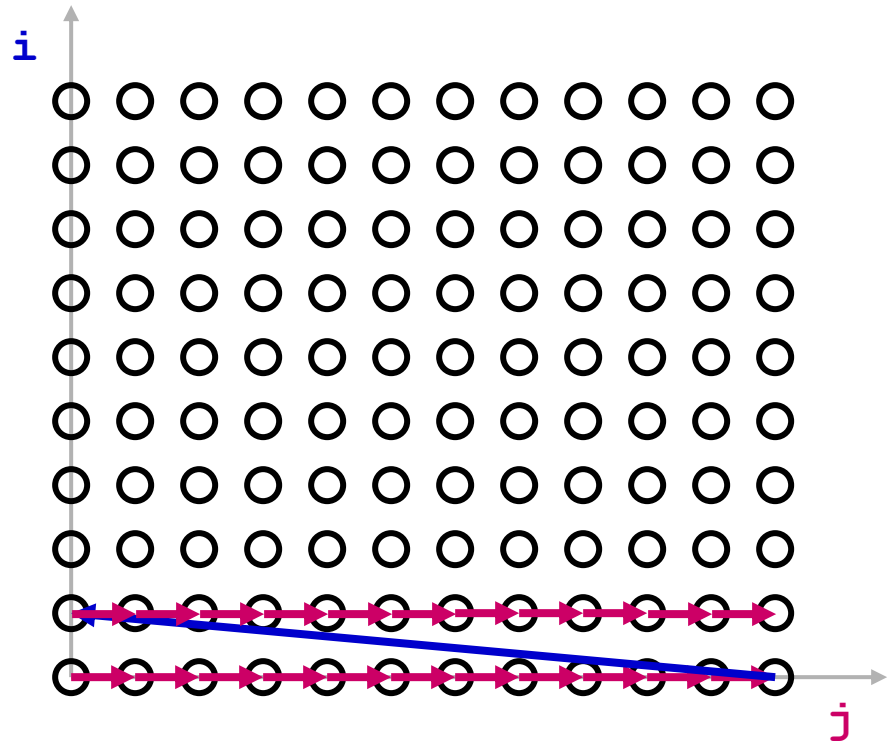
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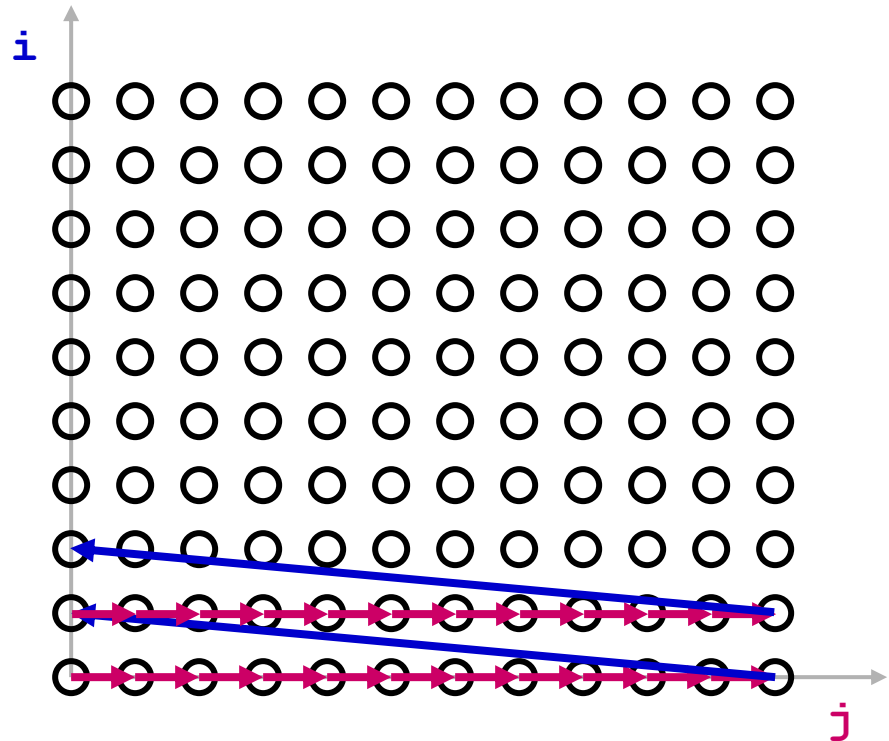
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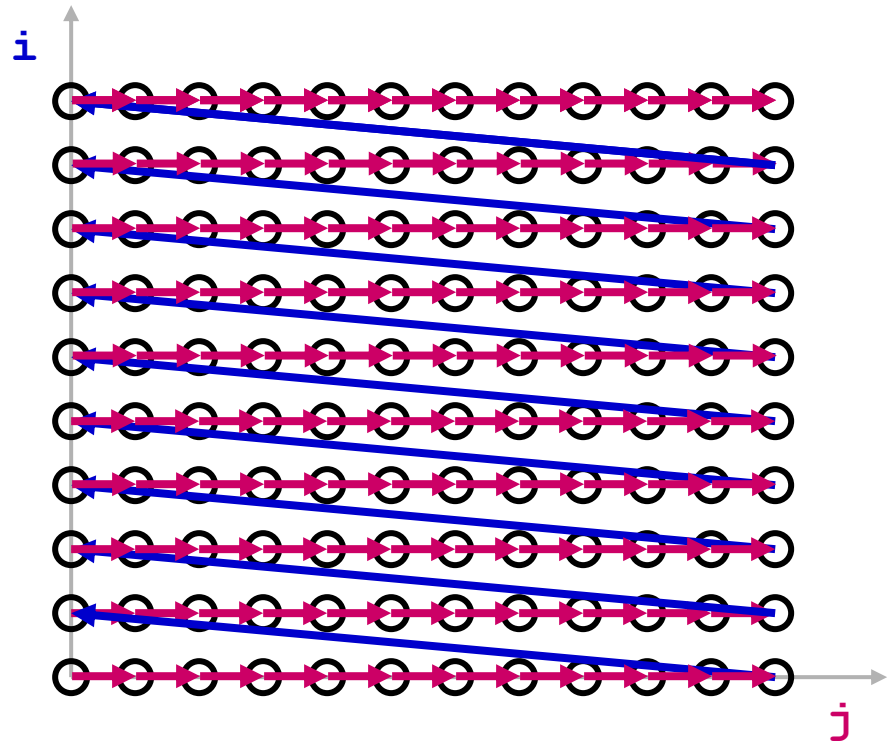
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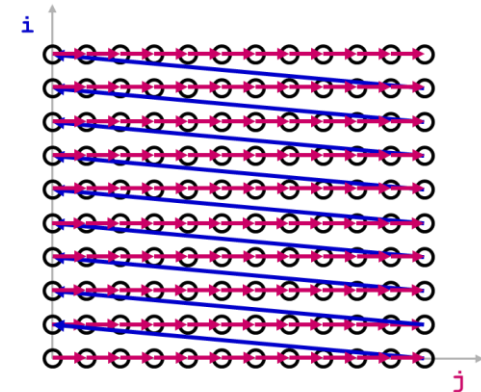


- Note: iteration space $\neq$ data space

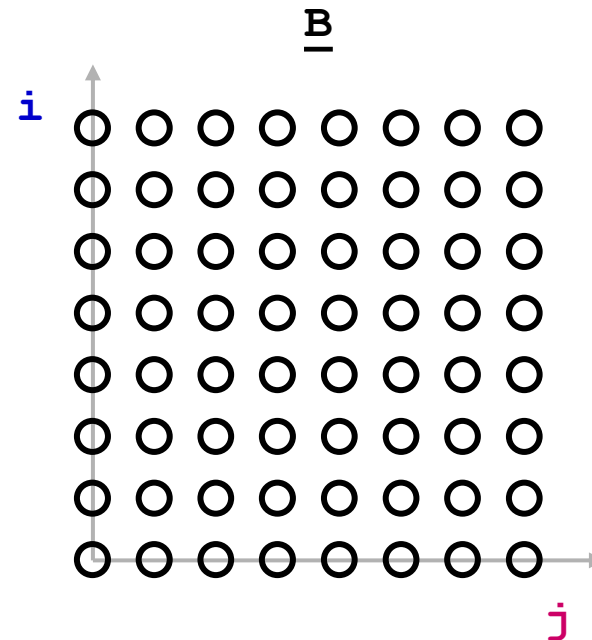
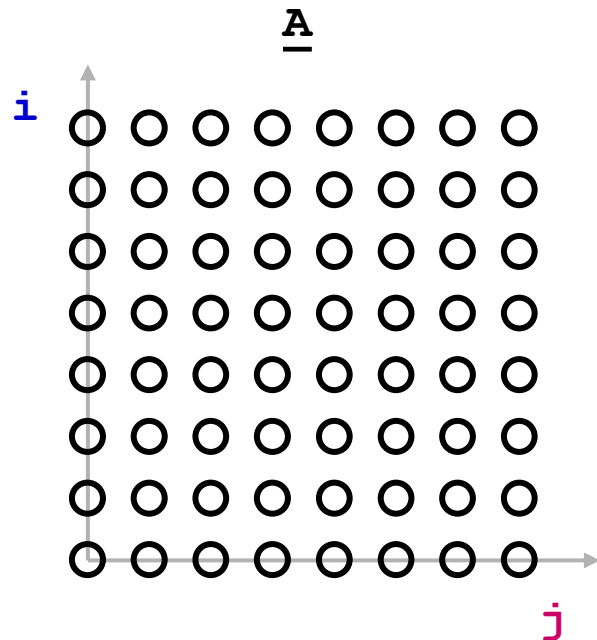
When Do Cache Misses Occur?

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ITERATION SPACE



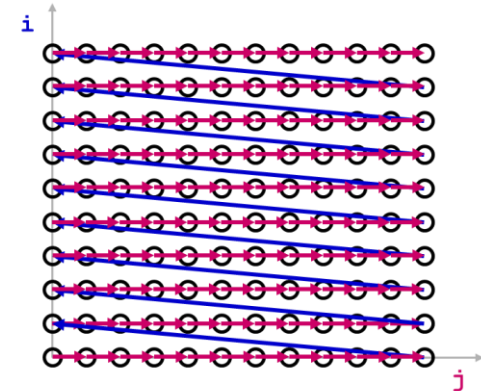
DATA SPACE



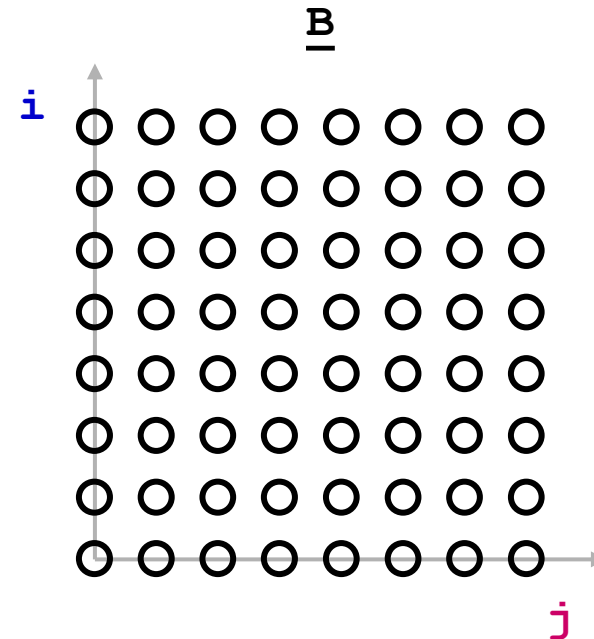
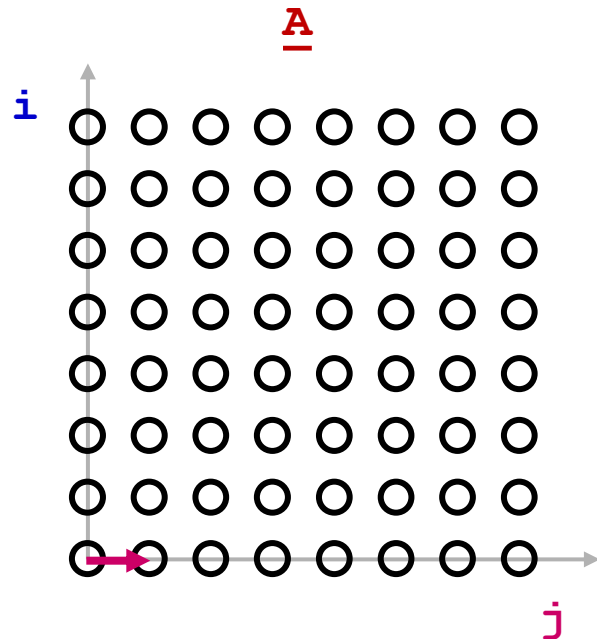
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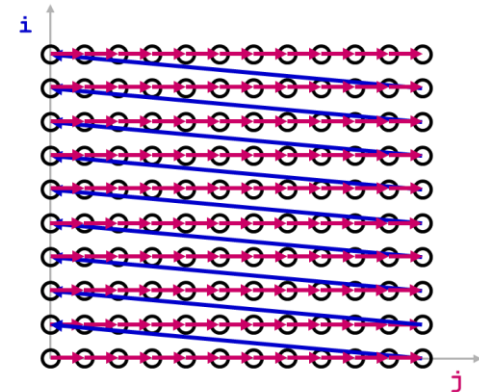
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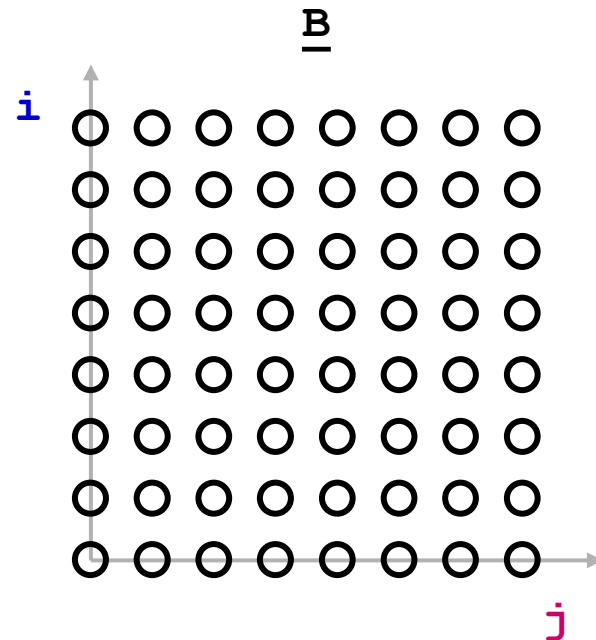
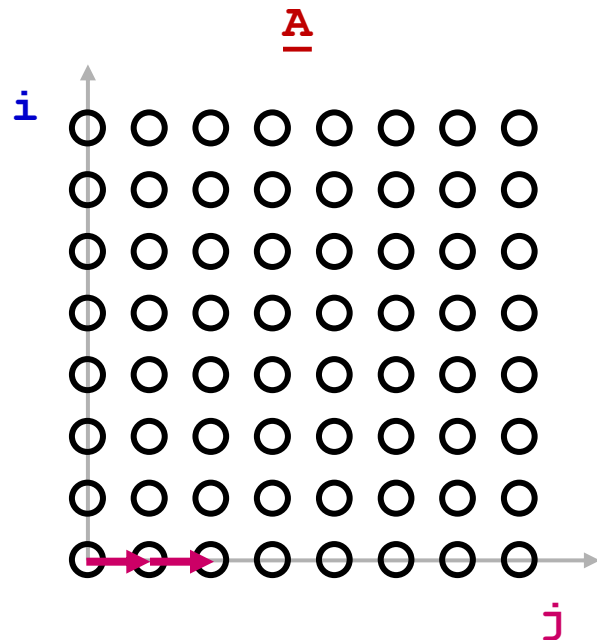
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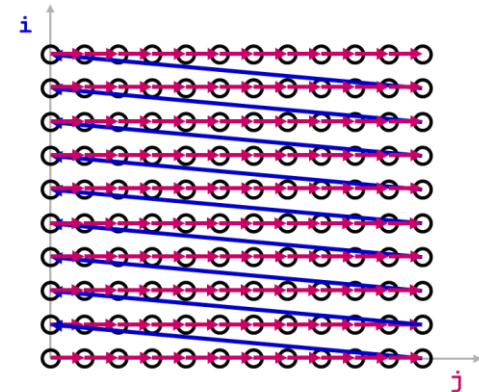
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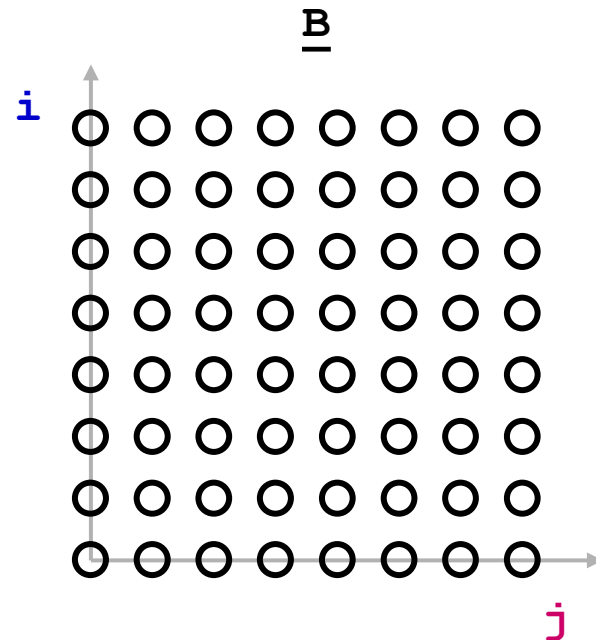
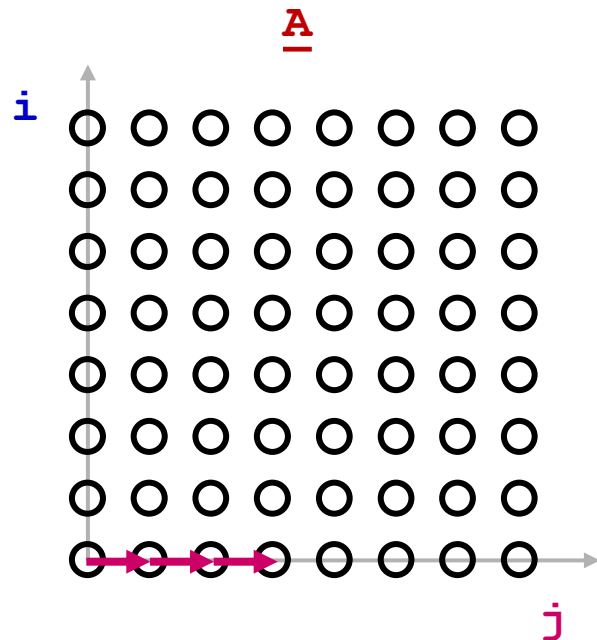
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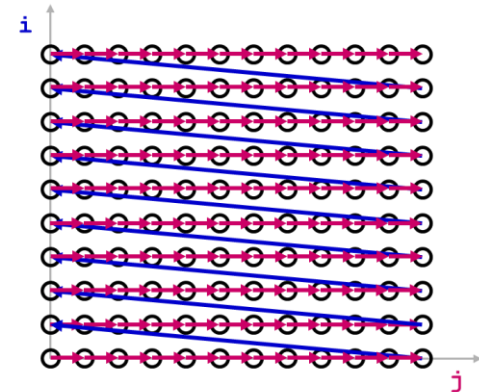
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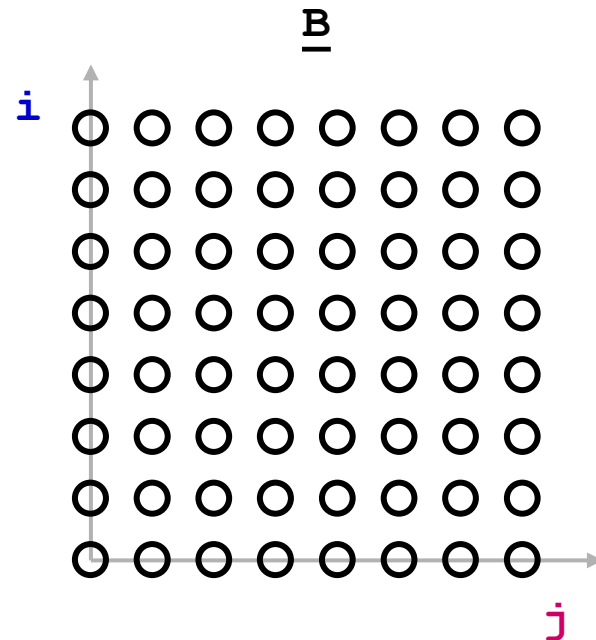
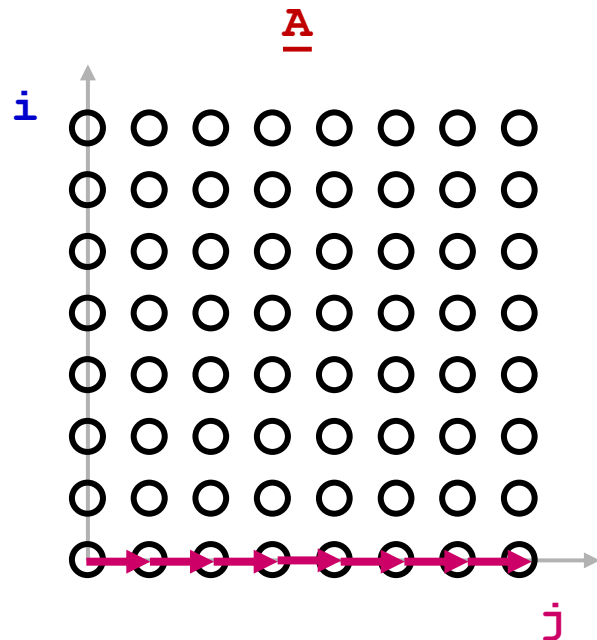
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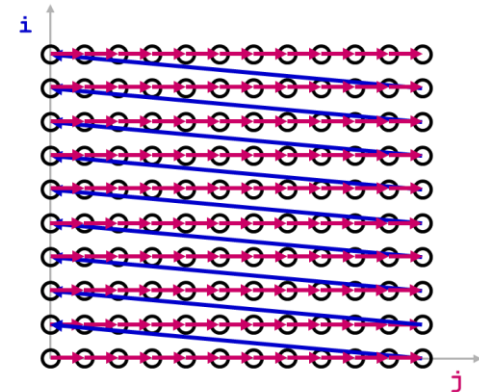
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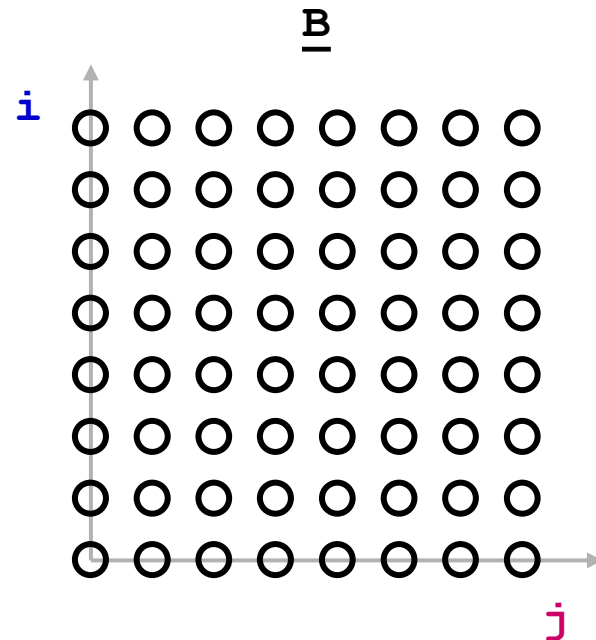
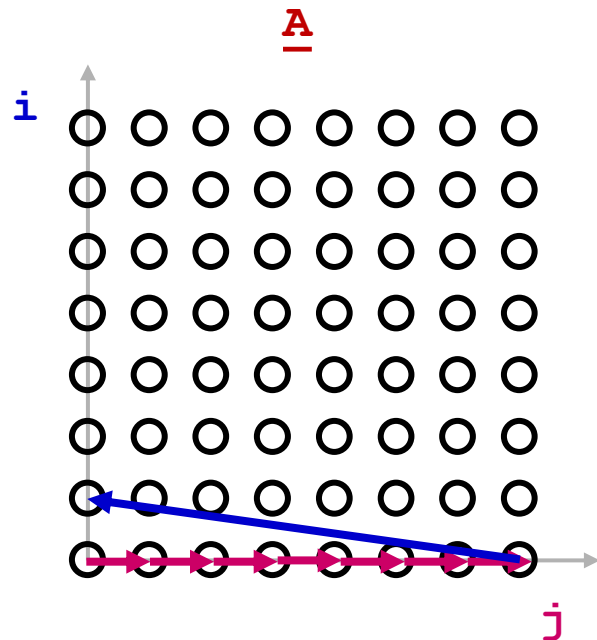
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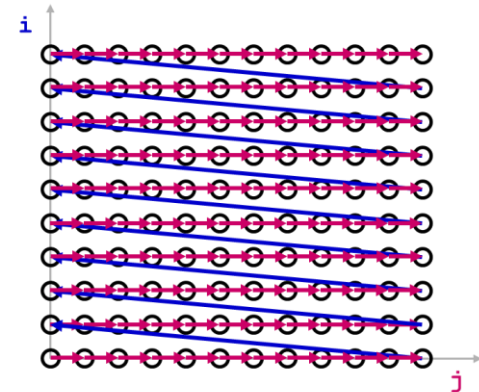
DATA SPACE



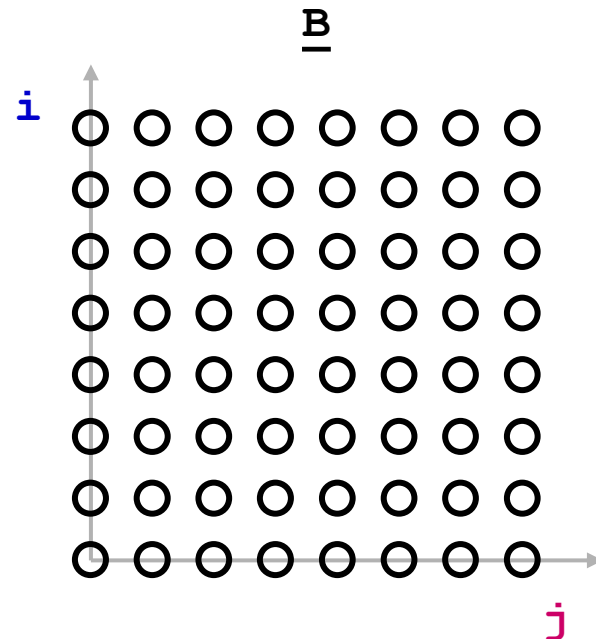
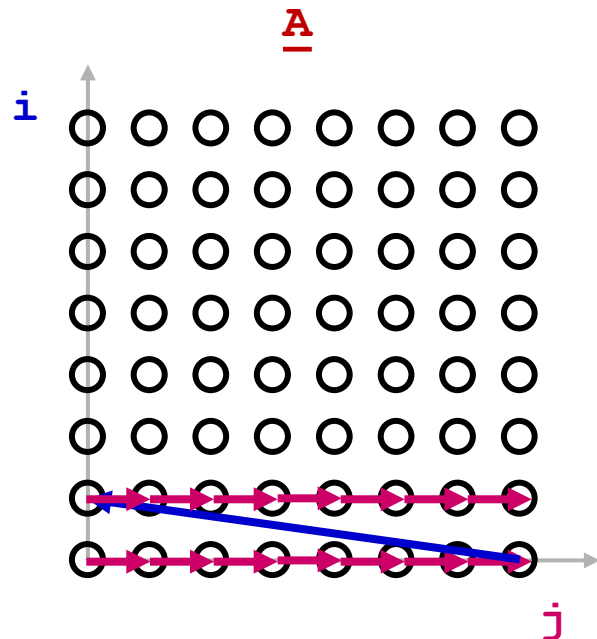
When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

ITERATION SPACE



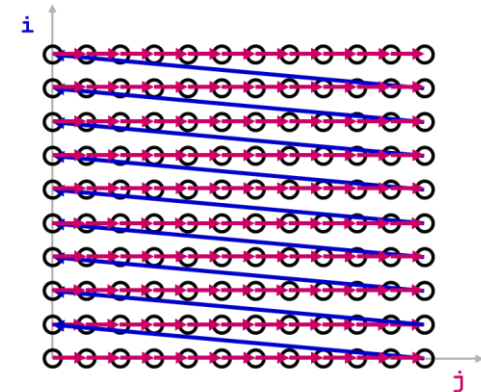
DATA SPACE



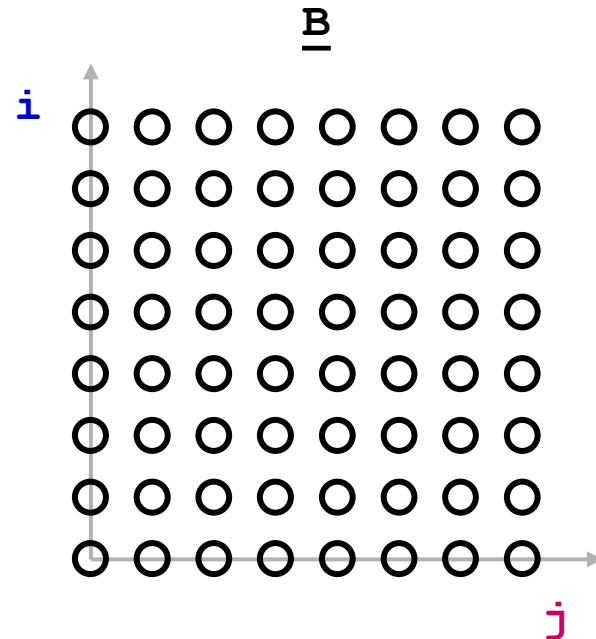
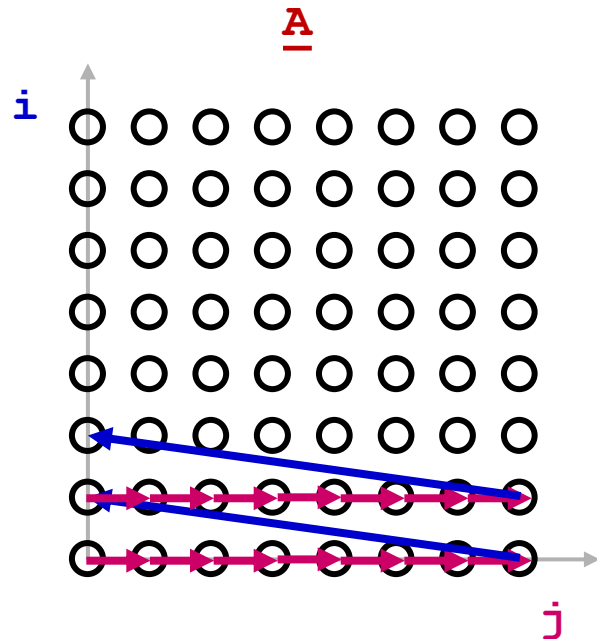
When Do Cache Misses Occur?

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    A[i][j] = B[j][i];
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ITERATION SPACE



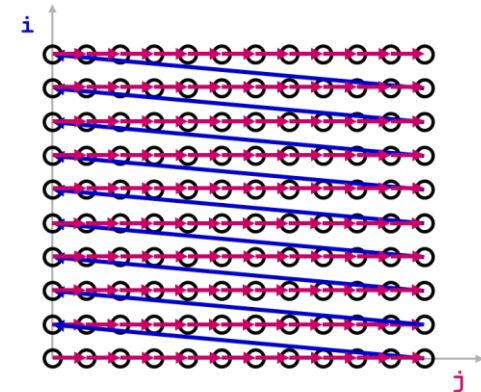
DATA SPACE



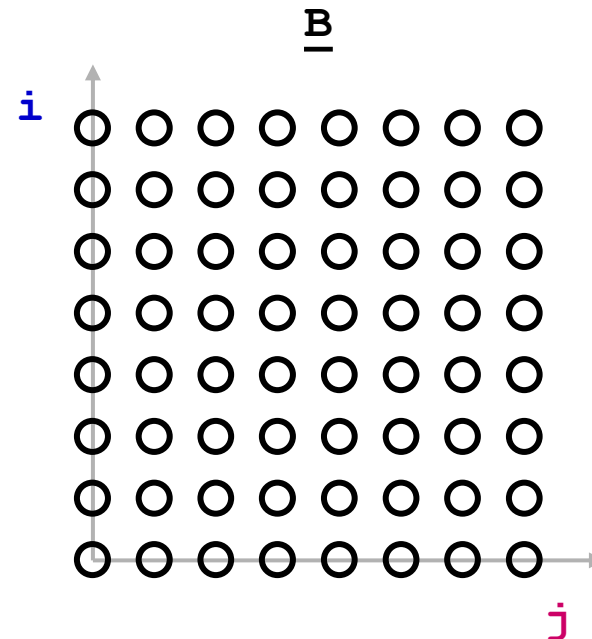
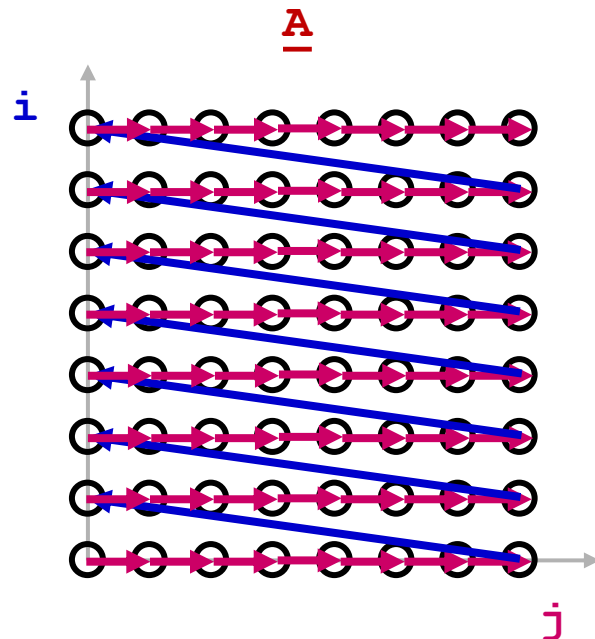
When Do Cache Misses Occur?

```
for i = 0 to N-1
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    A[i][j] = B[j][i];
```

ITERATION SPACE



DATA SPACE

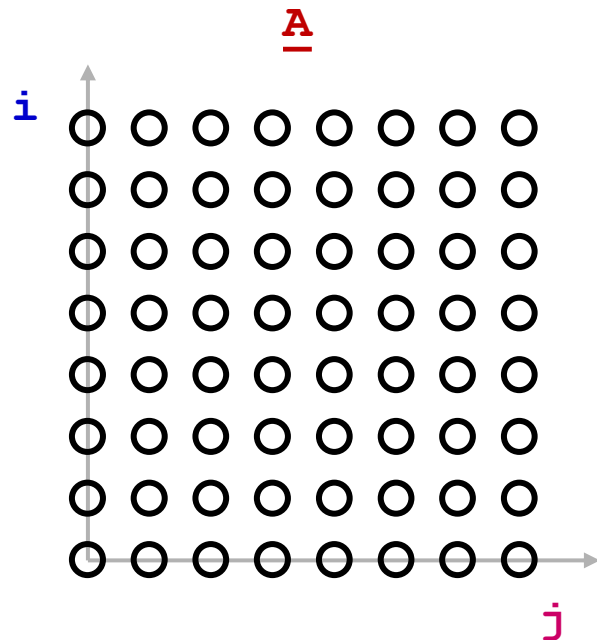


When Do Cache Misses Occur?

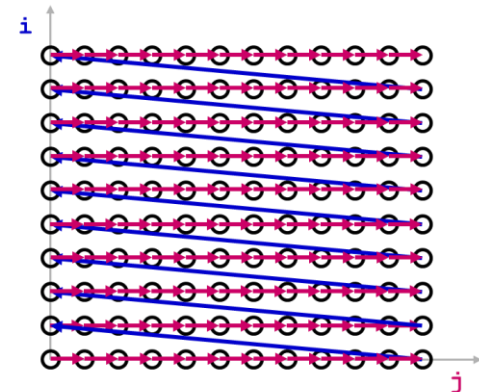
```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

When do cache misses occur?

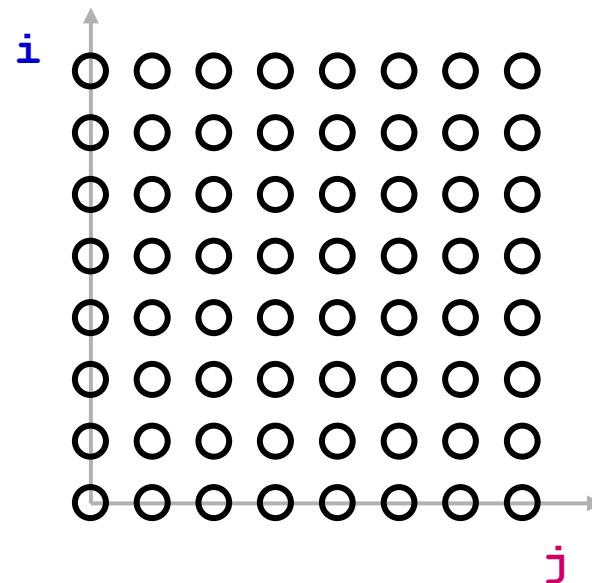
DATA SPACE



ITERATION SPACE



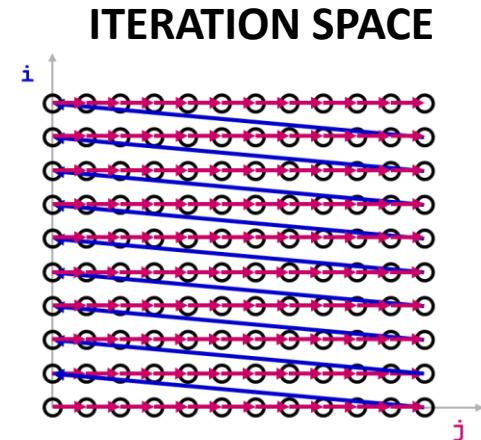
B



When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

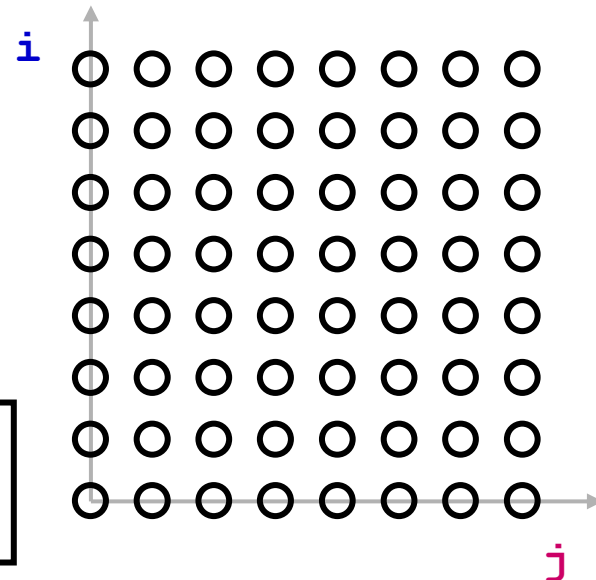
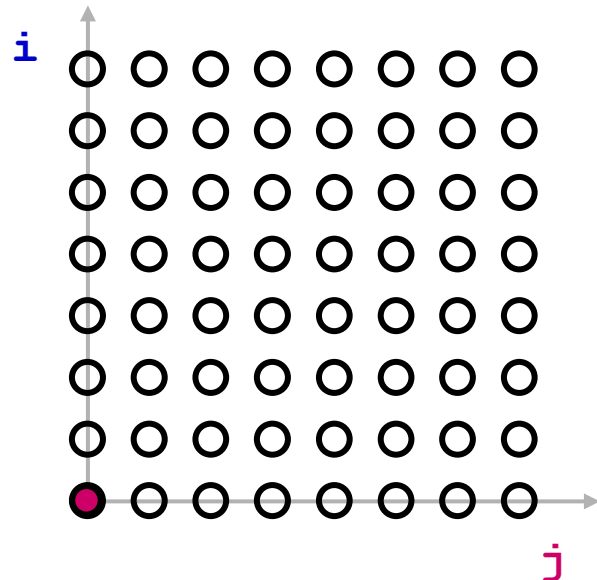
When do cache misses occur?



DATA SPACE

A Assuming 8 cache lines of 4 words

B



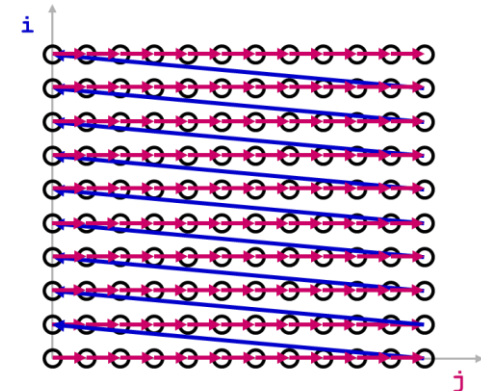
○ Hit
● Miss

When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

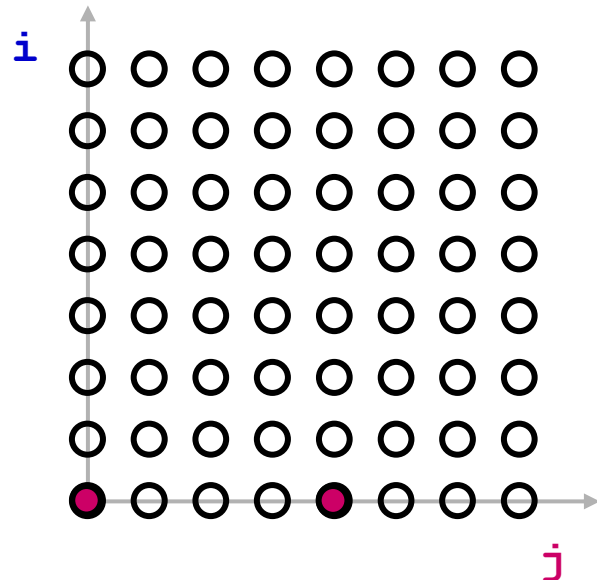
When do cache misses occur?

ITERATION SPACE

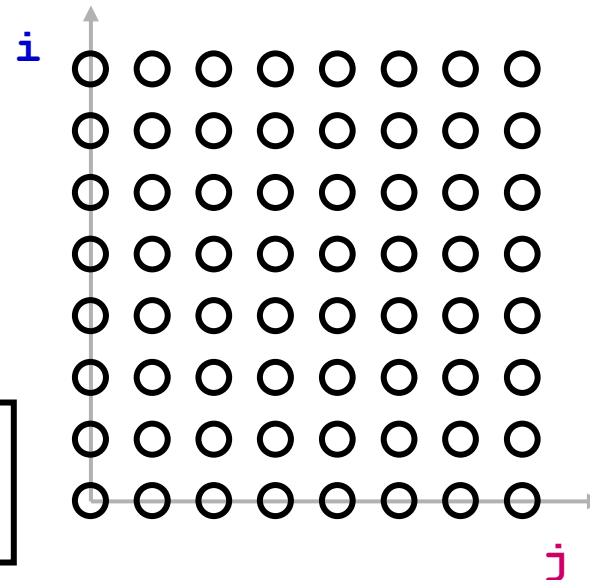


DATA SPACE

A Assuming 8 cache lines of 4 words



B



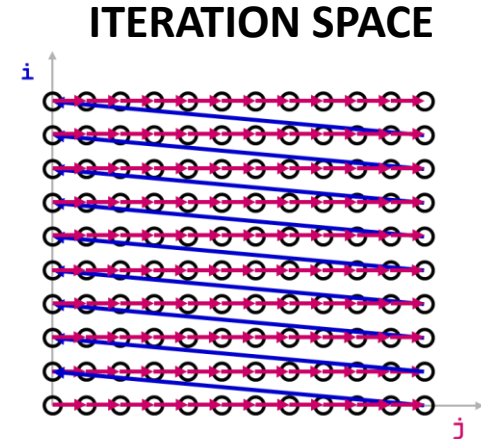
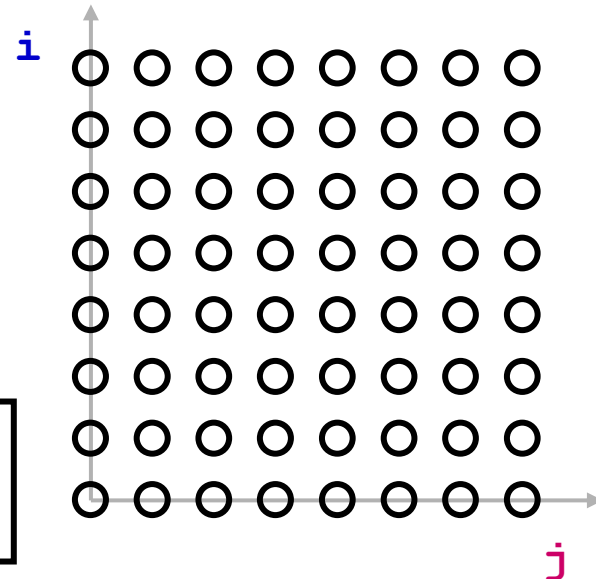
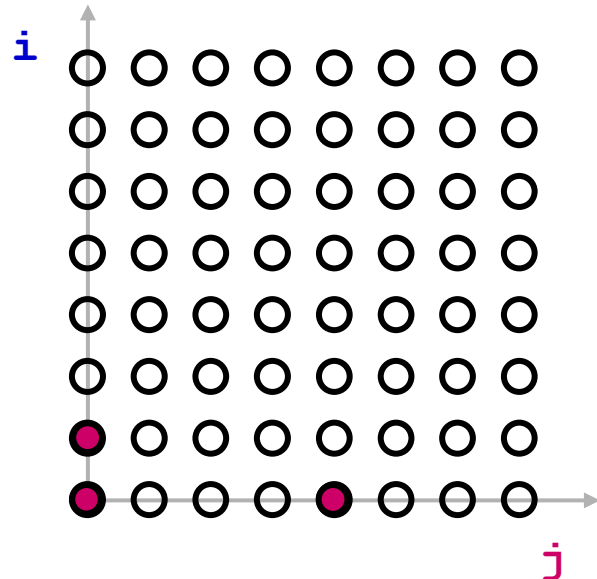
```
for i = 0 to N-1
    for j = 0 to N-1
        A[i][j] = B[j][i];
```

When do cache misses occur?

DATA SPACE

A Assuming 8 cache lines of 4 words

B



When Do Cache Misses Occur?

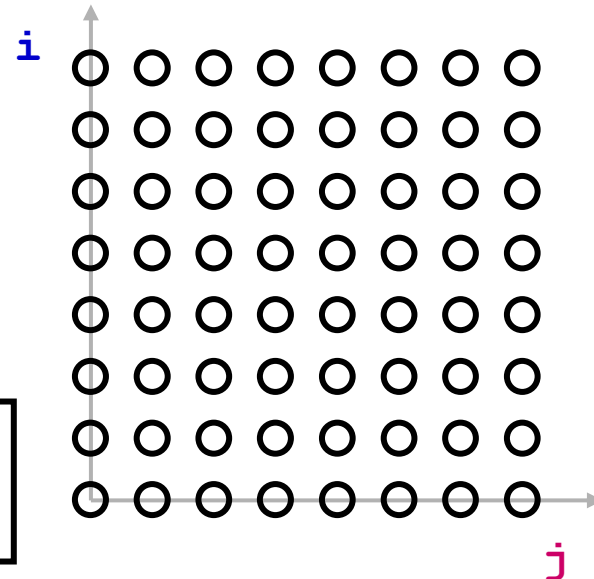
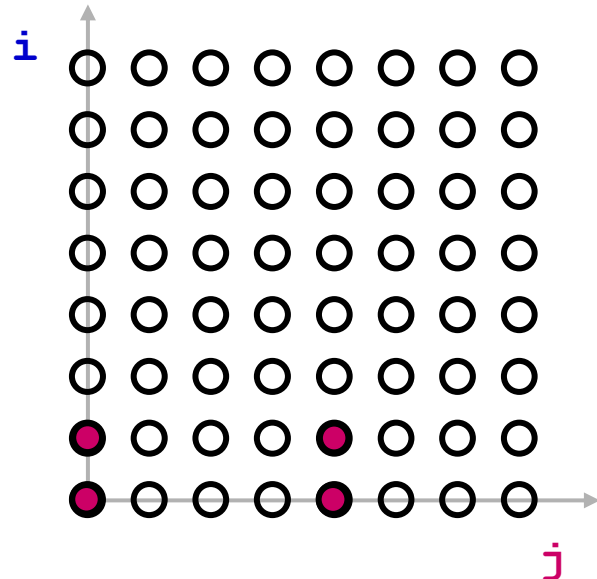
```
for i = 0 to N-1
    for j = 0 to N-1
        A[i][j] = B[j][i];
```

When do cache misses occur?

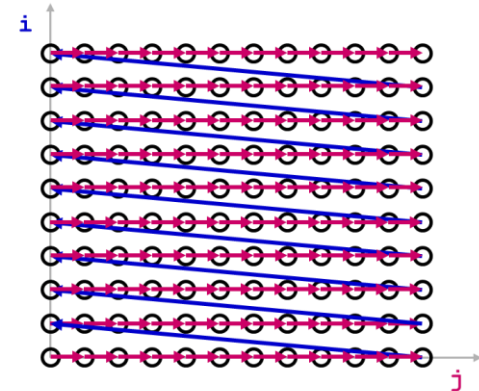
DATA SPACE

A Assuming 8 cache lines of 4 words

B



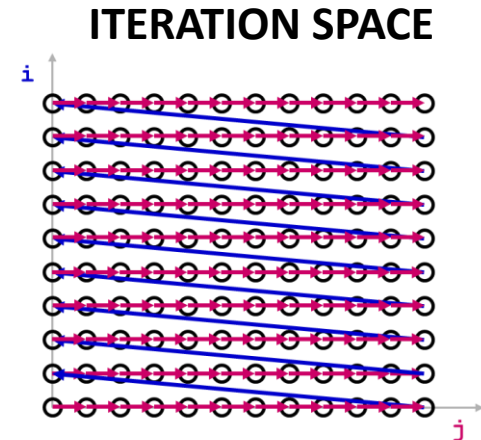
ITERATION SPACE



When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

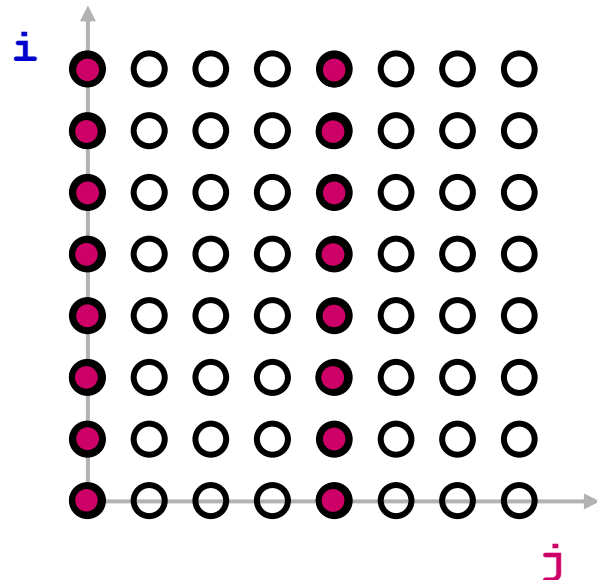
When do cache misses occur?



DATA SPACE

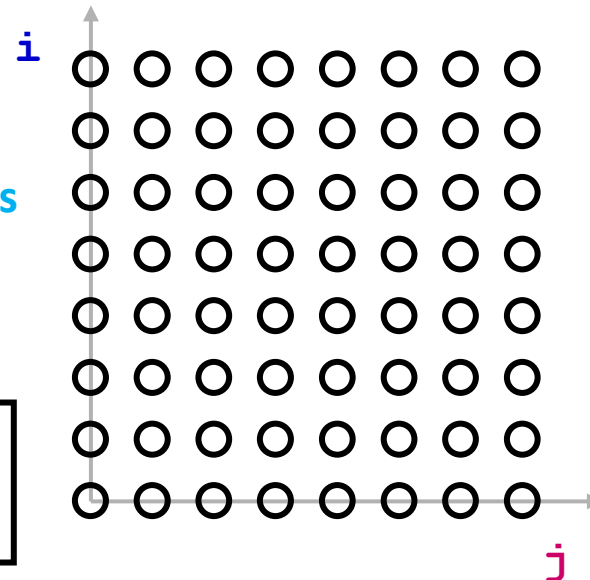
A Assuming 8 cache lines of 4 words

B



Which miss types?

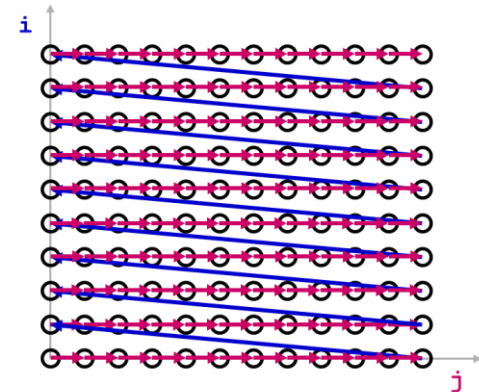
○ Hit
● Miss



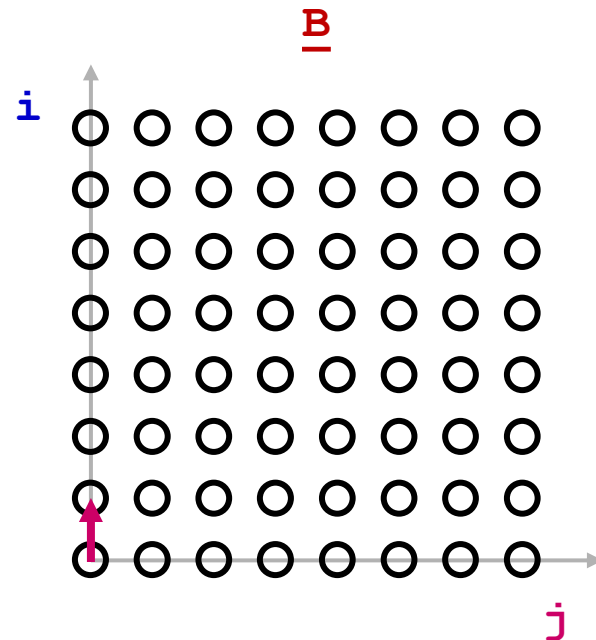
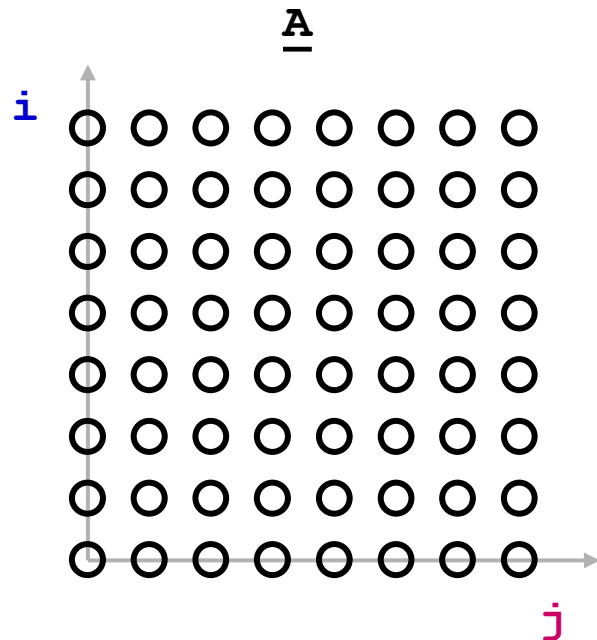
When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

ITERATION SPACE



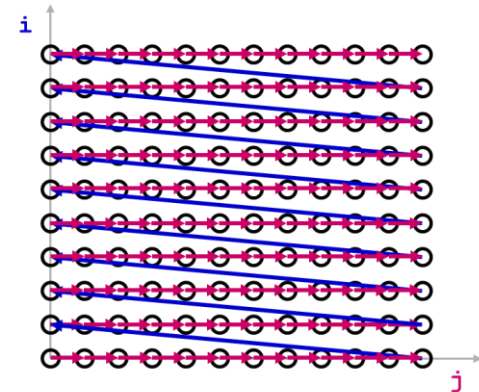
DATA SPACE



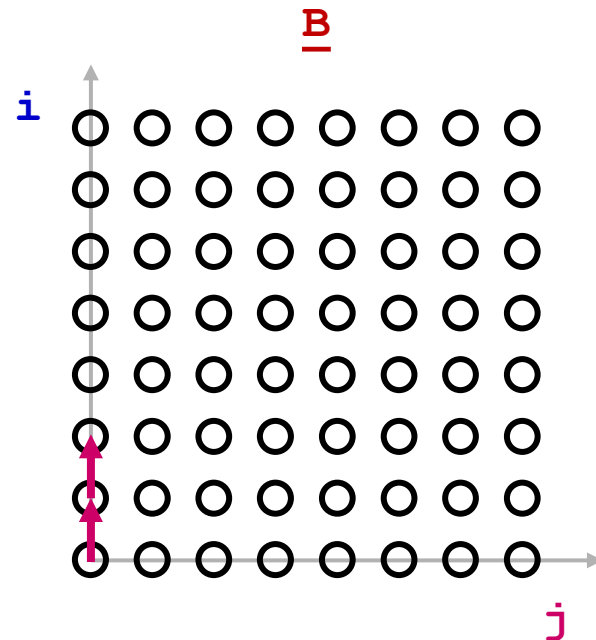
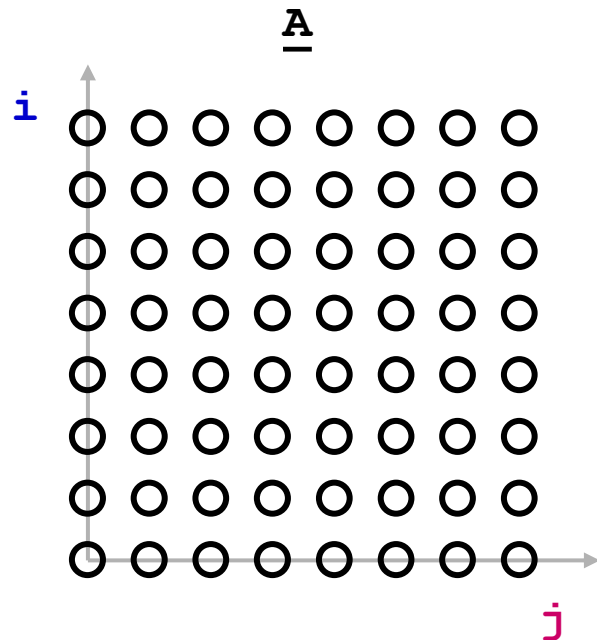
When Do Cache Misses Occur?

```
for i = 0 to N-1  
  for j = 0 to N-1  
    A[i][j] = B[j][i];
```

ITERATION SPACE



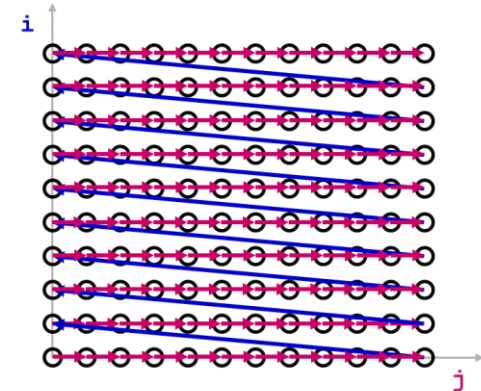
DATA SPACE



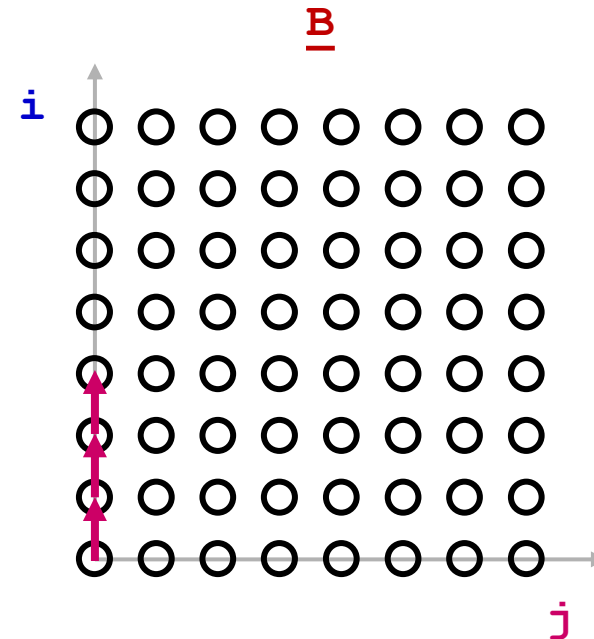
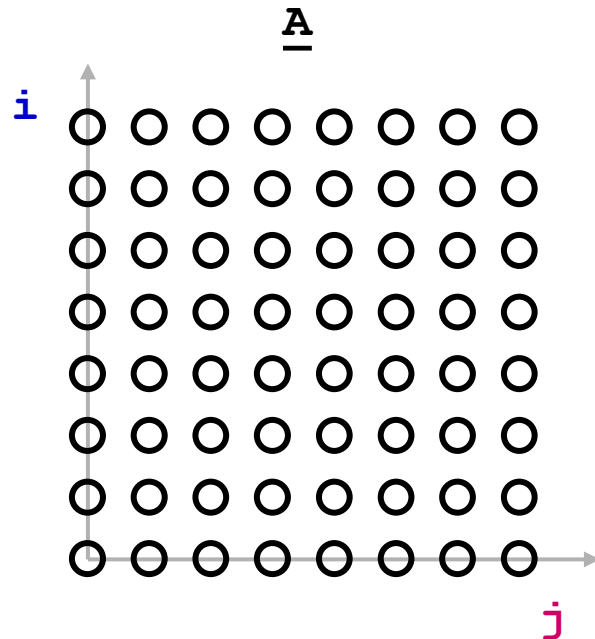
When Do Cache Misses Occur?

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for i = 0 to N-1
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```

ITERATION SPACE



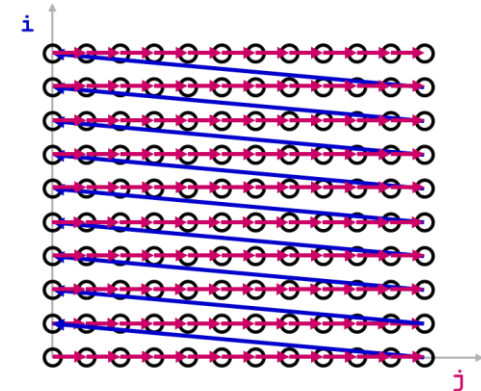
DATA SPACE



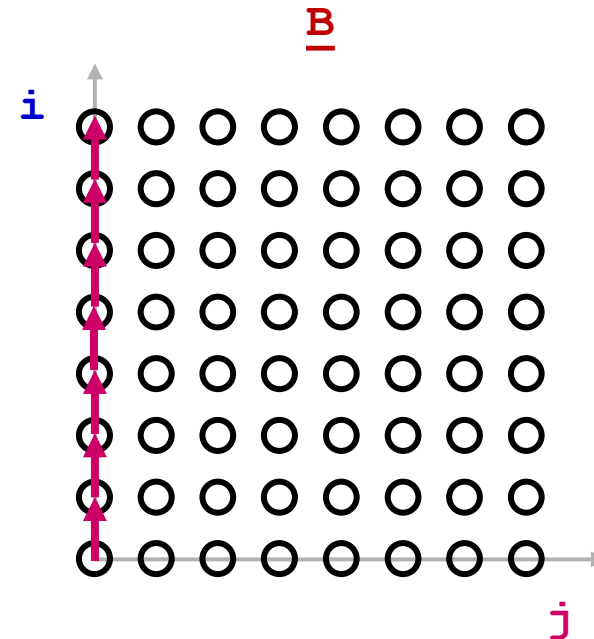
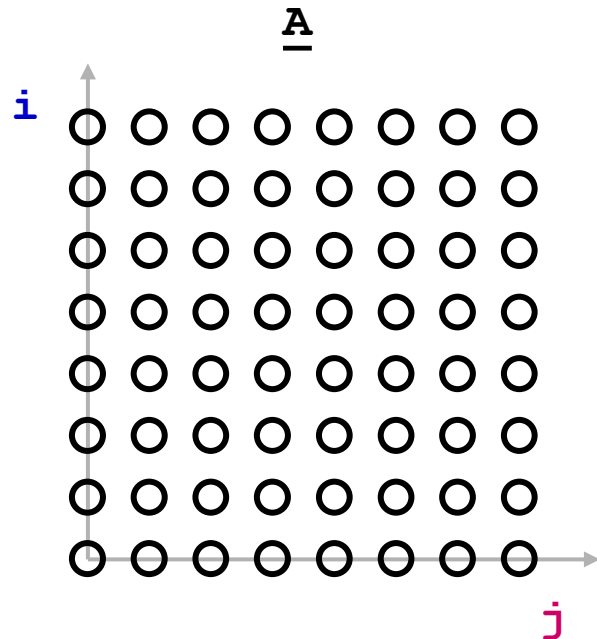
When Do Cache Misses Occur?

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```

ITERATION SPACE



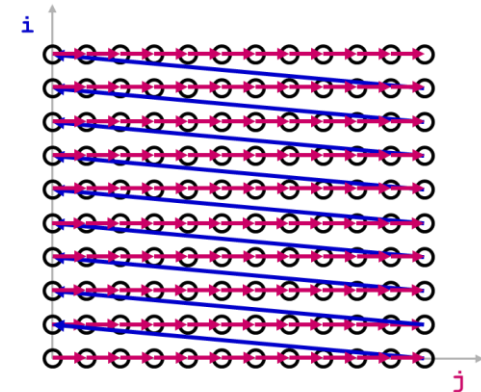
DATA SPACE



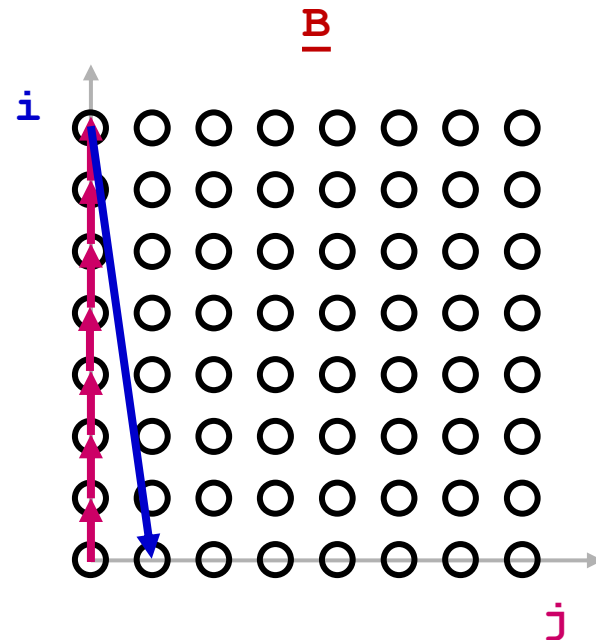
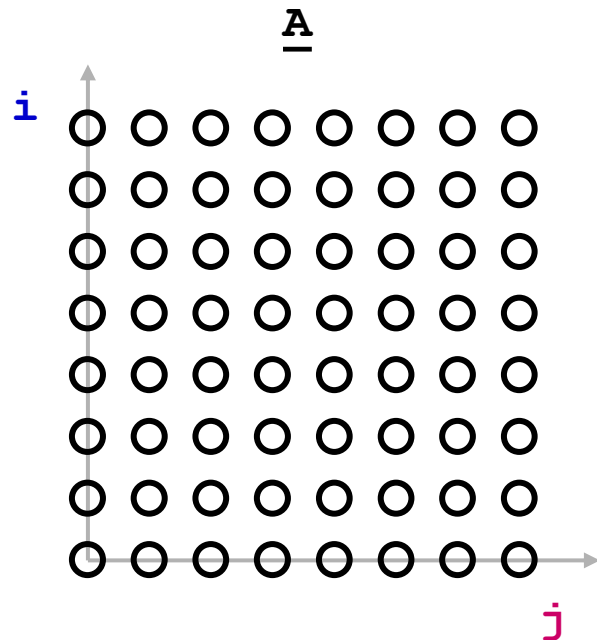
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ITERATION SPACE



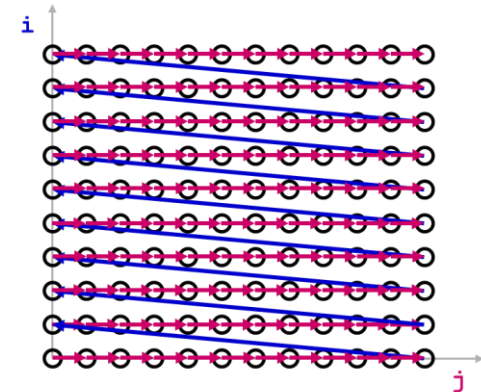
DATA SPACE



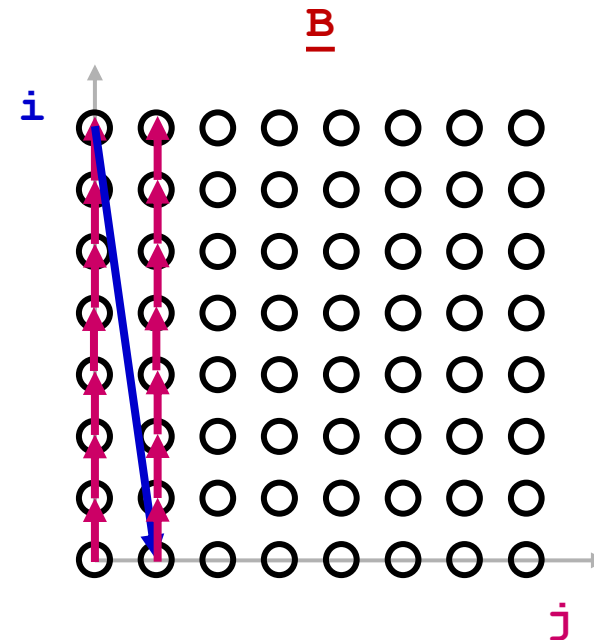
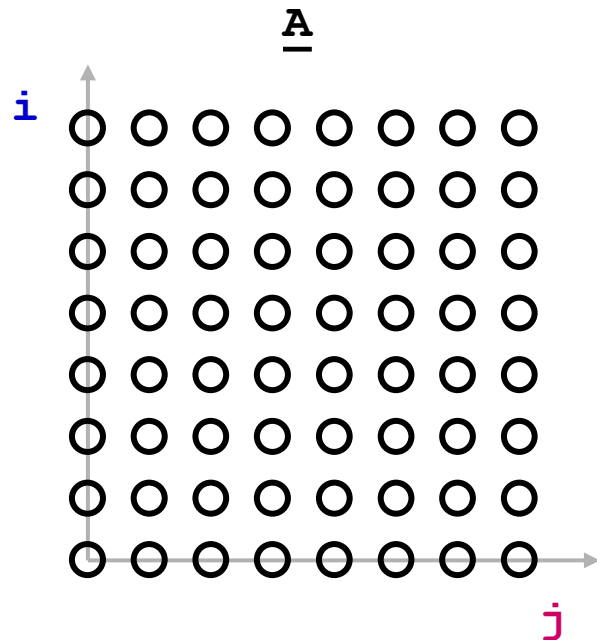
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ITERATION SPACE



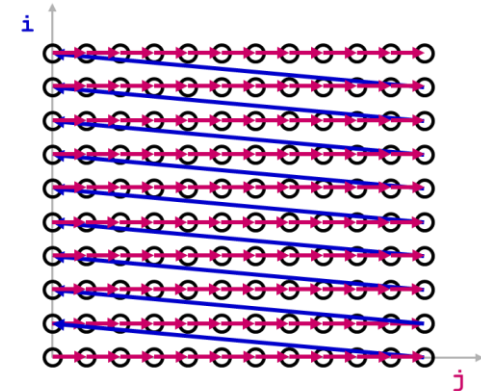
DATA SPACE



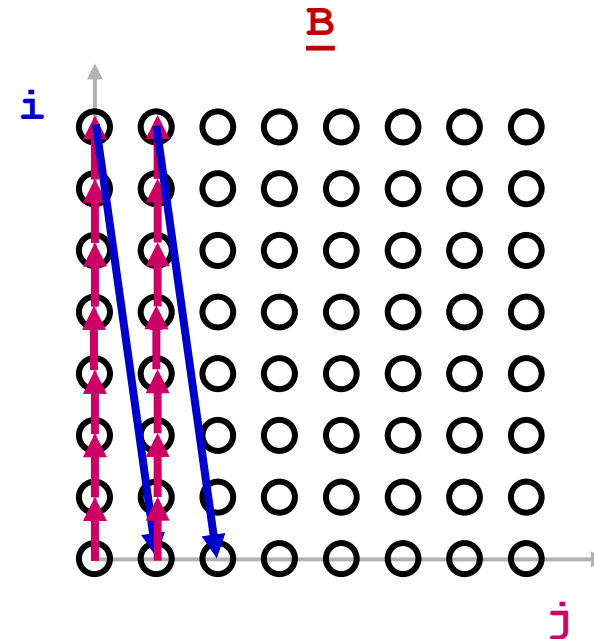
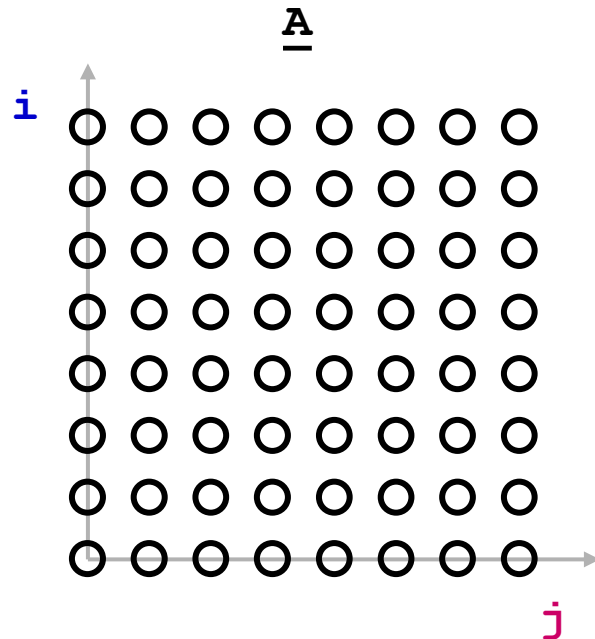
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ITERATION SPACE



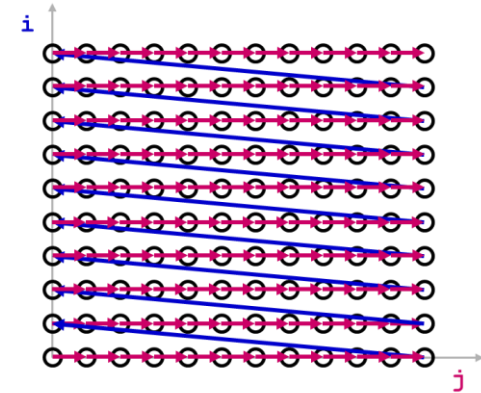
DATA SPACE



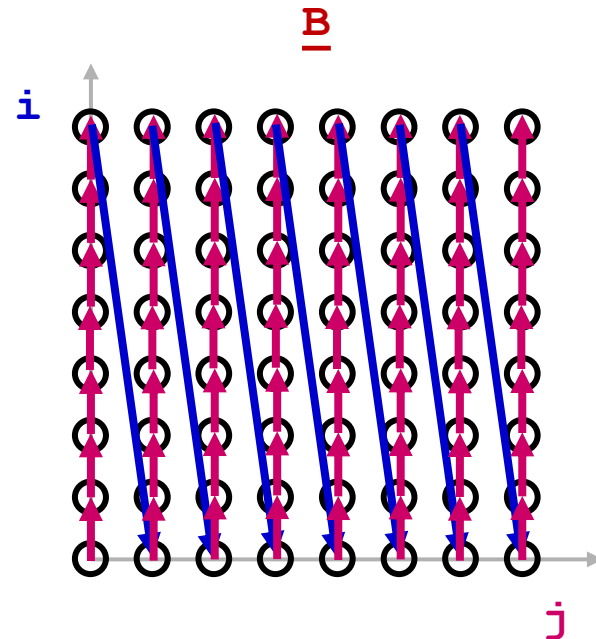
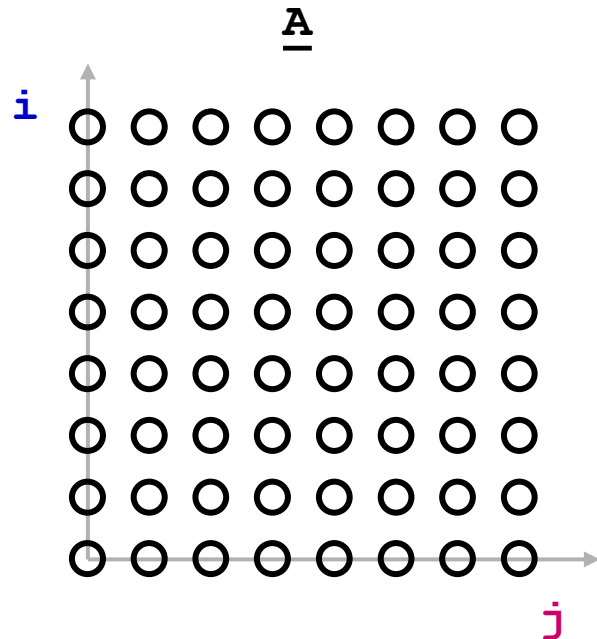
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ITERATION SPACE



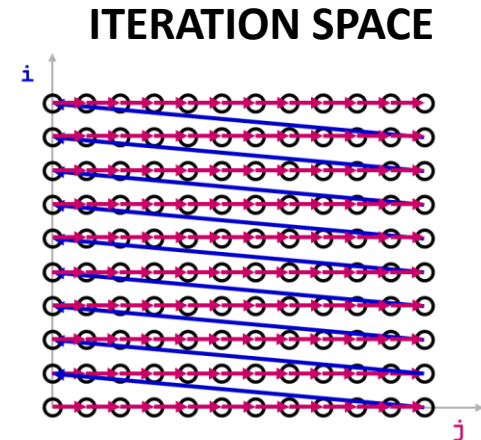
DATA SPACE



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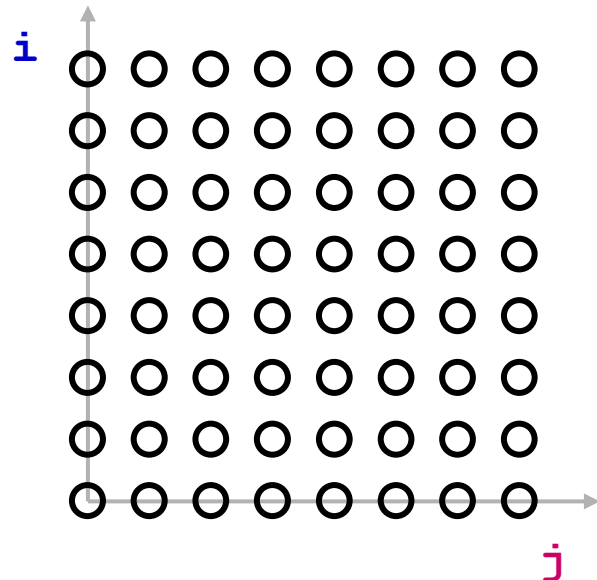
When do cache misses occur?



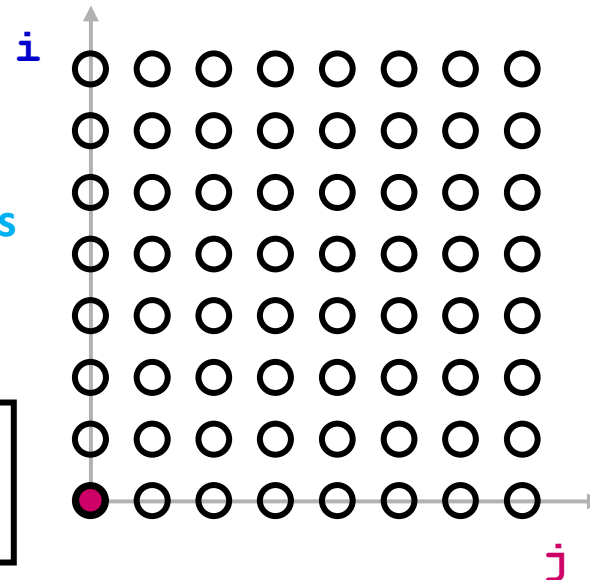
DATA SPACE

A Assuming 8 cache lines of 4 words

B



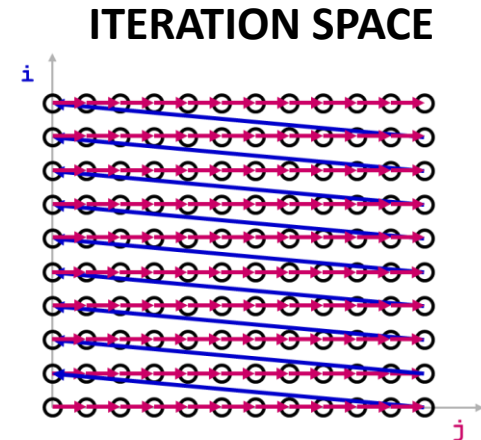
Which miss types?



When Do Cache Misses Occur?

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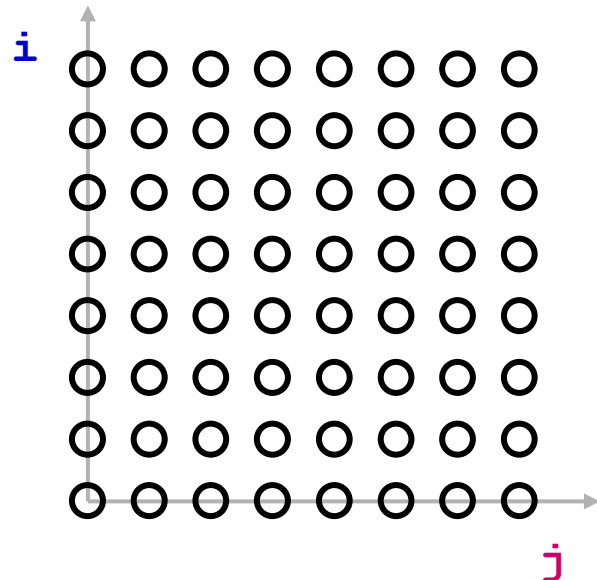
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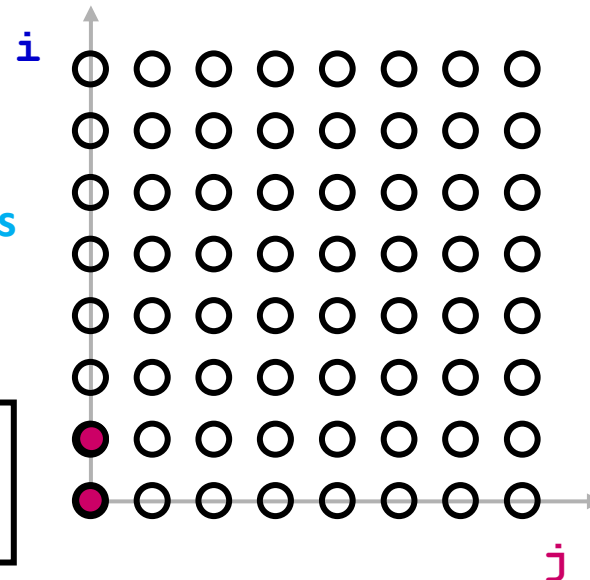
DATA SPACE

A Assuming 8 cache lines of 4 words

B



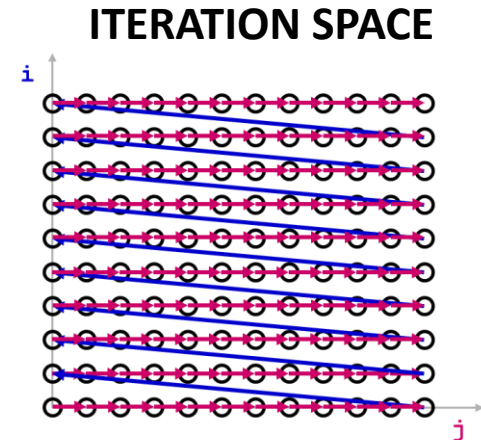
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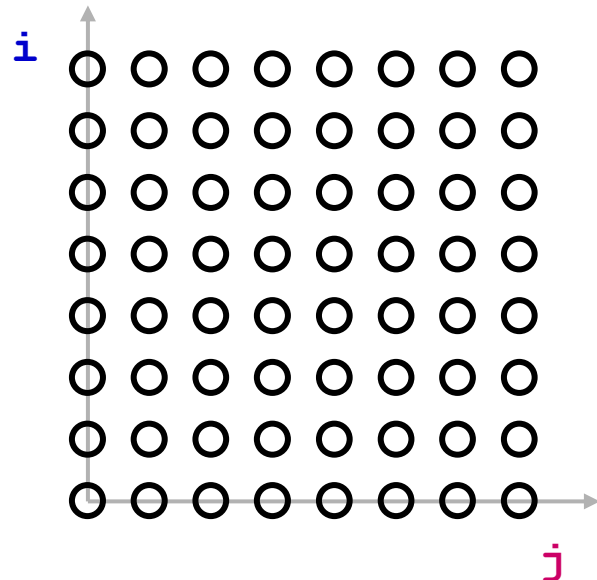
When do cache misses occur?



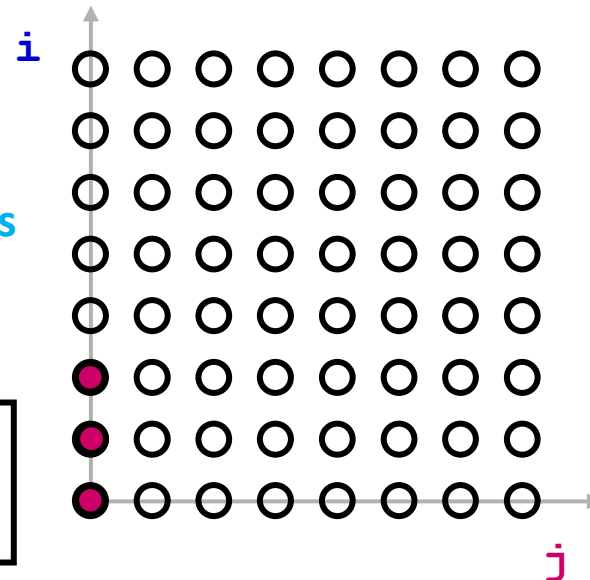
DATA SPACE

A Assuming 8 cache lines of 4 words

B



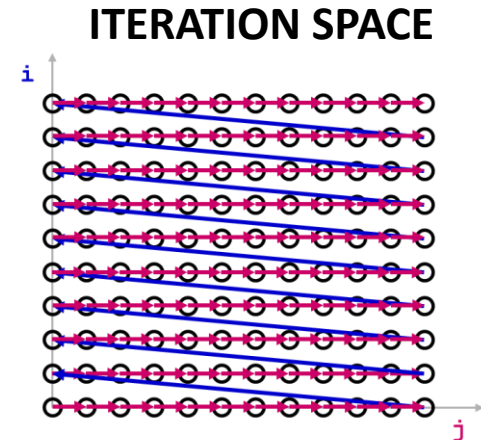
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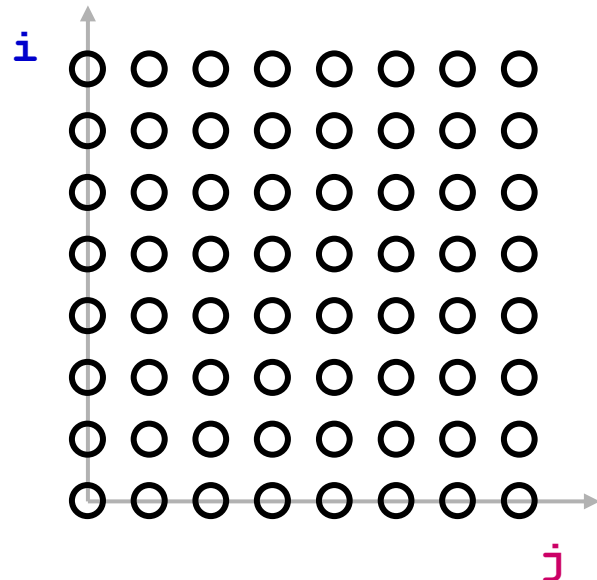
When do cache misses occur?



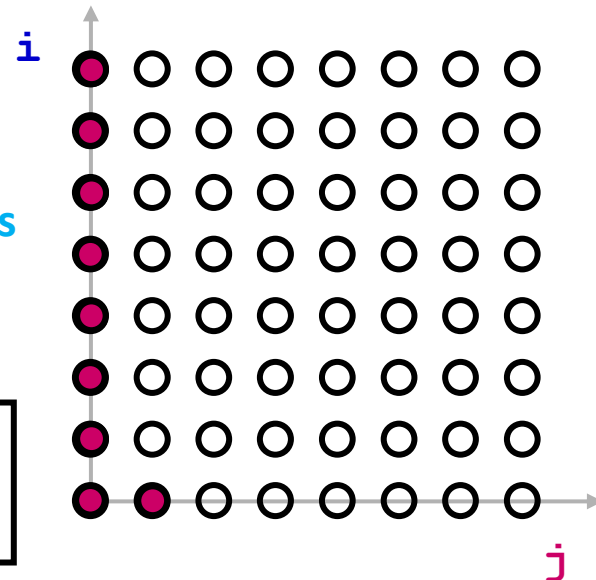
DATA SPACE

A Assuming 8 cache lines of 4 words

B



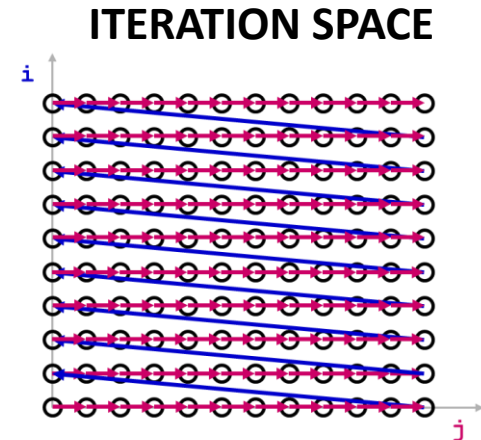
Which miss types?



When Do Cache Misses Occur?

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  for j = 0 to N-1
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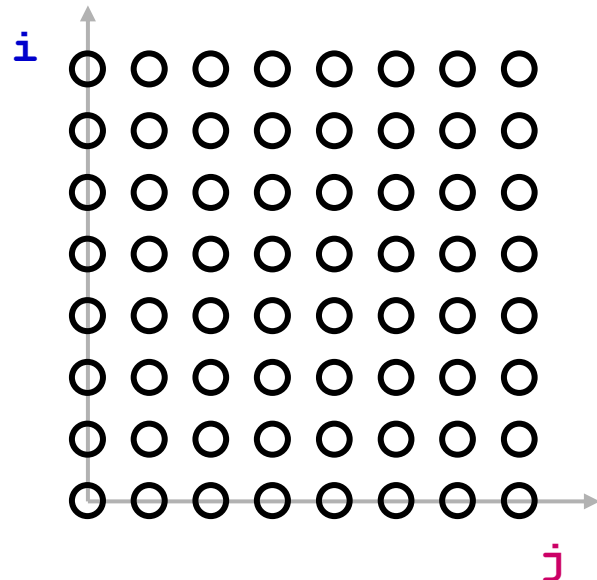
When do cache misses occur?



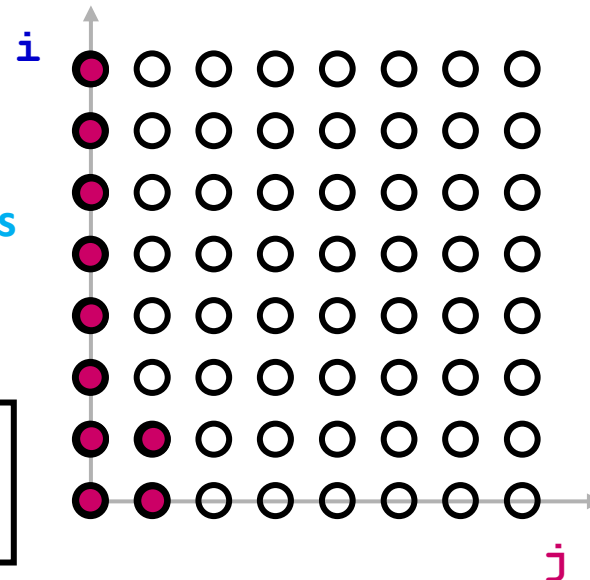
DATA SPACE

A Assuming 8 cache lines of 4 words

B



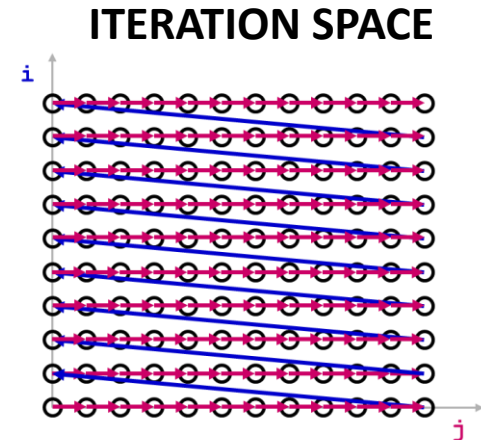
Which miss types?



When Do Cache Misses Occur?

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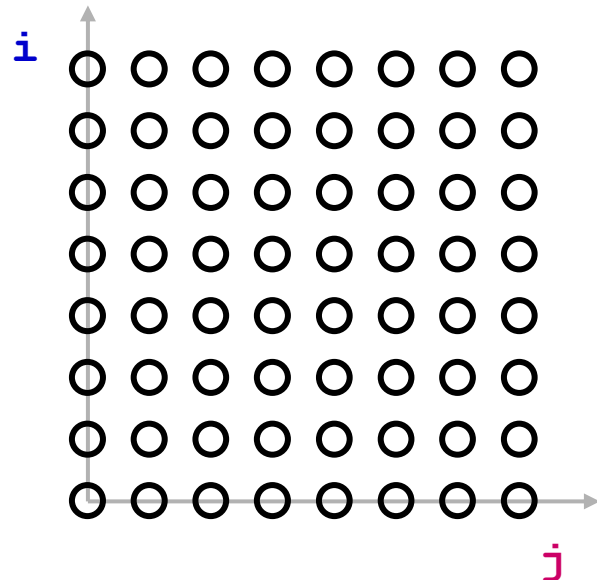
When do cache misses occur?



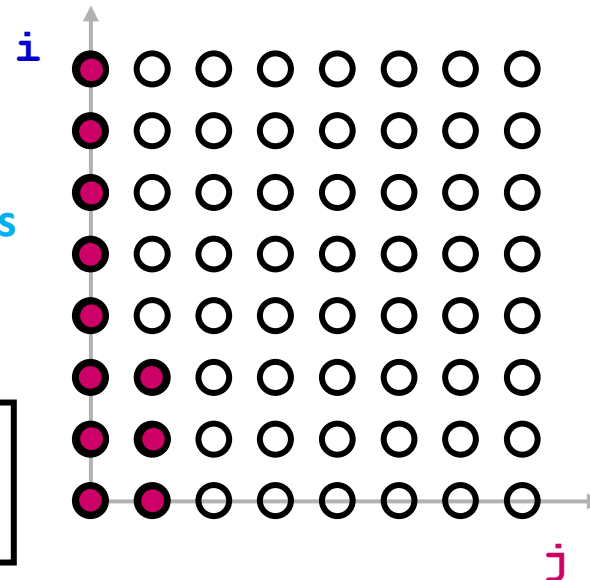
DATA SPACE

A Assuming 8 cache lines of 4 words

B



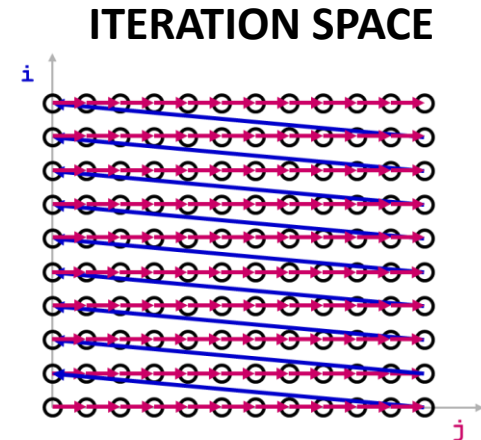
Which miss types?



When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

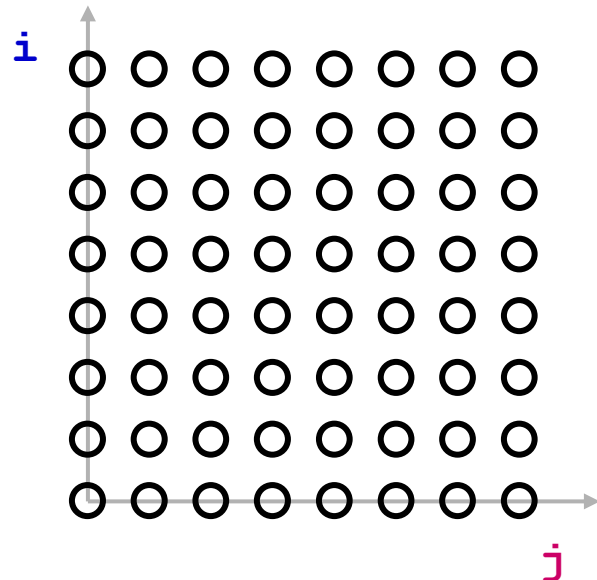
When do cache misses occur?



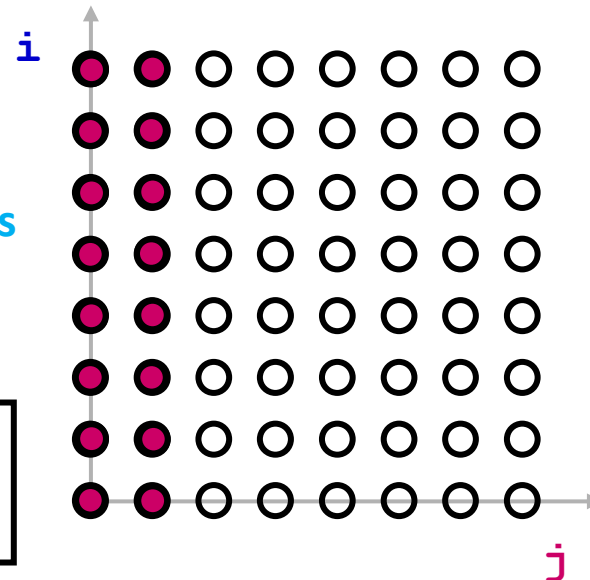
DATA SPACE

A Assuming 8 cache lines of 4 words

B



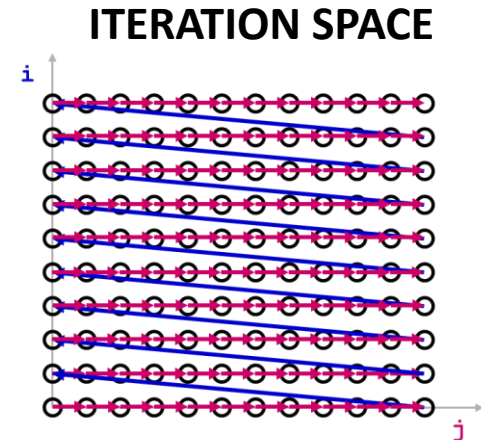
Which miss types?



When Do Cache Misses Occur?

```
for i = 0 to N-1
  for j = 0 to N-1
    A[i][j] = B[j][i];
```

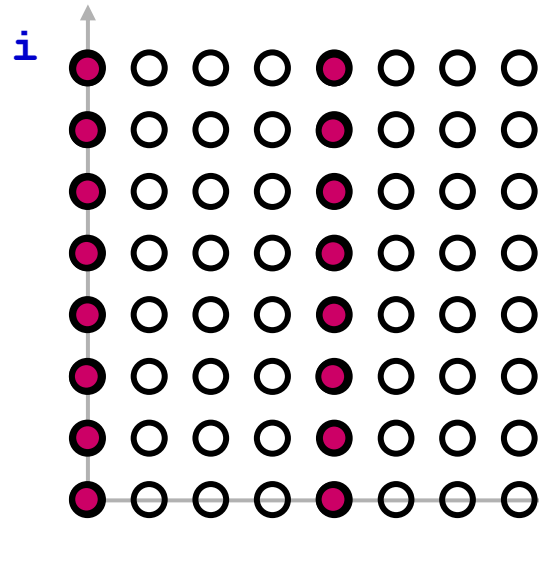
When do cache misses occur?



DATA SPACE

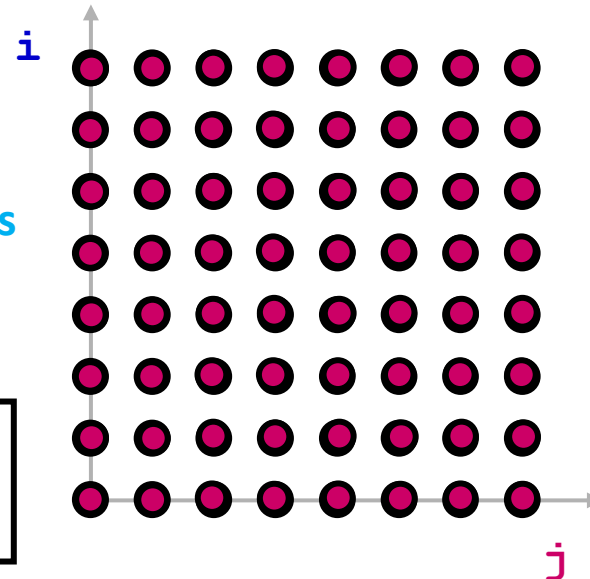
A Assuming 8 cache lines of 4 words

B



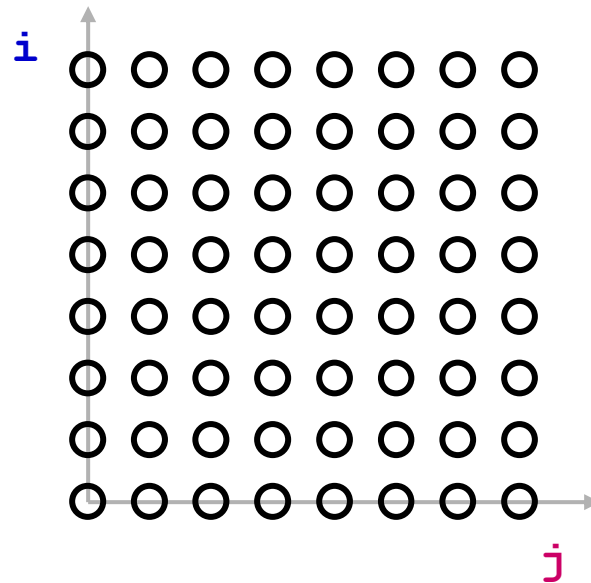
Which miss types?

○ Hit
● Miss



When Do Cache Misses Occur?

```
for i = 0 to N-1  
  for j = 0 to N-1  
    A[i+j][0] = i*j;
```



Cache Behavior of Array Accesses

- We need to answer the following questions:
 - when do cache misses occur?
 - use “locality analysis”
 - can we change the order of the iterations (or possibly data layout) to produce better behavior?
 - evaluate the cost of various alternatives
 - does the new ordering/layout still produce correct results?
 - use “dependence analysis”

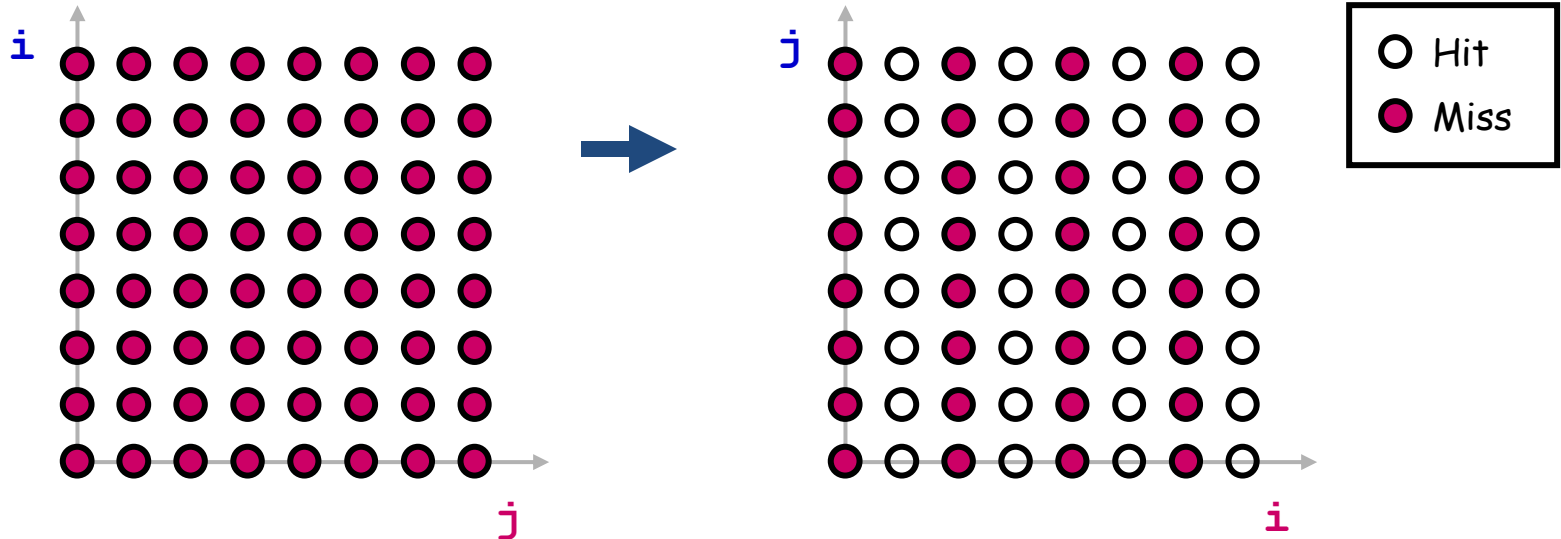
Examples of Loop Transformations

For improved cache behavior

- Loop Interchange
- Loop Tiling/Blocking
- ...

Loop Interchange

```
for i = 0 to N-1  
  for j = 0 to N-1  
    A[j][i] = i*j;  
for j = 0 to N-1  
  for i = 0 to N-1  
    A[j][i] = i*j;
```



- (assuming N is large relative to cache size)

Permutation has many goals

- Locality optimization
 - Particularly, for spatial locality
- Rearrange loop nest to move parallelism to appropriate level of granularity
 - Inward to exploit fine-grain parallelism
 - Outward to exploit coarse-grain parallelism
- Also, to enable other optimizations

Cache Blocking in Two Dimensions

Matrix multiplication

```
for i = 0 to N-1
  for j = 0 to N-1
    for k = 0 to N-1
      c[i][j] += a[i][k]*b[k][j];
```

- Imagine a situation in which your data structure does not fit in cache, but has lots of reuse
- The traversal order is key to exploiting temporal (and spatial) locality

Cache Blocking in Two Dimensions

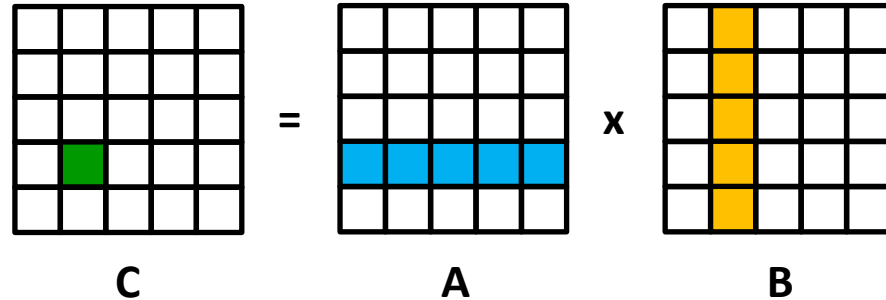
Matrix multiplication

```
for i = 0 to N-1
```

```
  for j = 0 to N-1
```

```
    for k = 0 to N-1
```

```
      c[i][j] += a[i][k]*b[k][j];
```



1 element = cache line size = S

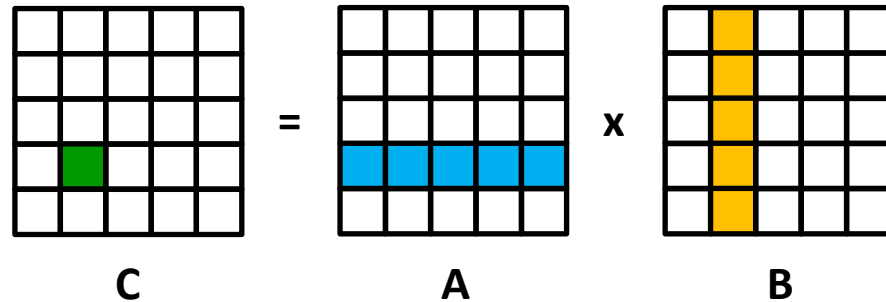
N elements = 1 matrix row does not fit in cache

- **In the innermost loop:**
- $c[i][j]$ is accessed only once – no reuse
- $a[i][k]$ entails accessing a whole A matrix row – doesn't fit in cache
- $b[k][j]$ entails accessing a whole B matrix column – doesn't fit in cache and has poor spatial locality

Cache Blocking in Two Dimensions

Matrix multiplication

```
for i = 0 to N-1
  for j = 0 to N-1
    for k = 0 to N-1
      c[i][j] += a[i][k]*b[k][j];
```



- **How many cache misses?**
- **C – NxN** – Each row element is written N times in sequence (the element stays in cache)
- **A – NxNxN** – Each row is read N times in sequence (but a row doesn't fit in cache)
- **B – NxNxN** – Each matrix element is read once in column-major order

Cache Blocking in Two Dimensions

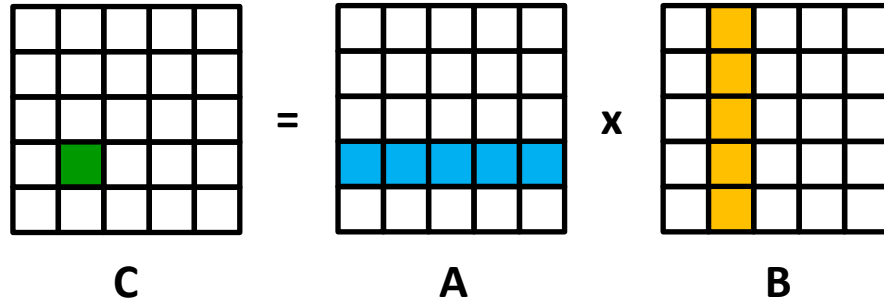
Matrix multiplication

```
for i = 0 to N-1
```

```
  for j = 0 to N-1
```

```
    for k = 0 to N-1
```

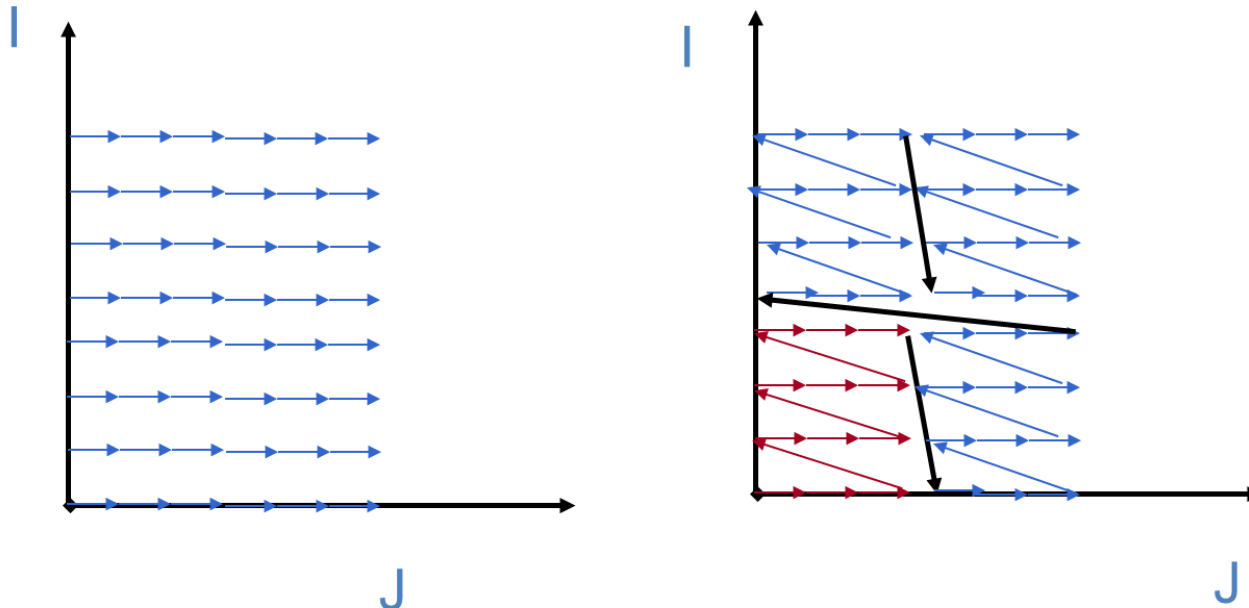
```
      c[i][j] += a[i][k]*b[k][j];
```



- **How many cache misses?** - $2N^3 + N^2$
- C – $N \times N$ – Each row element is written N times in sequence (the element stays in cache)
- A – $N \times N \times N$ – Each row is read N times in sequence (but a row doesn't fit in cache)
- B – $N \times N \times N$ – Each matrix element is read once in column-major order

Cache Blocking in Two Dimensions

- Blocking reorders loop iterations to bring iterations that reuse data closer in time
- Goal is to retain in cache between reuse



Cache Blocking in Two Dimensions

Matrix multiplication

```
for (jj=0; jj<N; jj+=B)
  for (kk=0; kk<N; kk+=B)
    for i = 0 to N-1
      for (j = jj; j < jj+B; j++)
        for (k = kk; k < kk+B; k++)
          c[i][j] += a[i][k]*b[k][j];
```

```
for i = 0 to N-1
  for j = 0 to N-1
    for k = 0 to N-1
      c[i][j] += a[i][k]*b[k][j];
```

- Cache blocking alters the traversal order of the matrices involved

Cache Blocking in Two Dimensions

Matrix multiplication

```
for (jj=0; jj<N; jj+=B)
  for (kk=0; kk<N; kk+=B)
    for i = 0 to N-1
      for (j = jj; j < jj+B; j++)
        for (k = kk; k < kk+B; k++)
          c[i][j] += a[i][k]*b[k][j];
```

```
for i = 0 to N-1
  for j = 0 to N-1
    for k = 0 to N-1
      c[i][j] += a[i][k]*b[k][j];
```

- Cache blocking alters the traversal order of the matrices involved
- How many misses?

Tiling is Fundamental!

- Tiling is very commonly used to manage limited storage
 - Registers
 - Caches
 - Software-managed buffers
 - Small main memory
- Can be applied hierarchically

Determining Safety and Profitability

- Safety
 - Notion of reordering transformations
 - Based on data dependences
- Profitability
 - Reuse analysis (and other cost models)
 - Also based on data dependences, but simpler

Loop fusion

- Some programs have separate loops that access same array(s)
 - If the arrays are large the data is replaced before it is reused in the second loop
- **Loop fusion:** "fuse" loops into a single loop
 - Data fetched in the cache can be reused before being replaced
 - Reduces miss rate by improving temporal locality

Loop fusion

- Example

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        a[i][j] = 1/b[i][j]*c[i][j];
```

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        d[i][j] = a[i][j]+c[i][j];
```

Loop fusion

- Example

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        a[i][j] = 1/b[i][j]*c[i][j];
```

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        d[i][j] = a[i][j]+c[i][j];
```

Loop fusion

- Example

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        a[i][j] = 1/b[i][j]*c[i][j];
```

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        d[i][j] = a[i][j]+c[i][j];
```

```
for (i=0; i<N; i++)  
    for (j=0; j<N; j++)  
        a[i][j] = 1/b[i][j]*c[i][j];  
        d[i][j] = a[i][j]+c[i][j];
```

Loop fusion

- When is loop fusion allowed?
- Changing execution order should not violate dependencies



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Software Prefetching

Coping with Memory Latency

Reduce Latency:

— Locality Optimizations

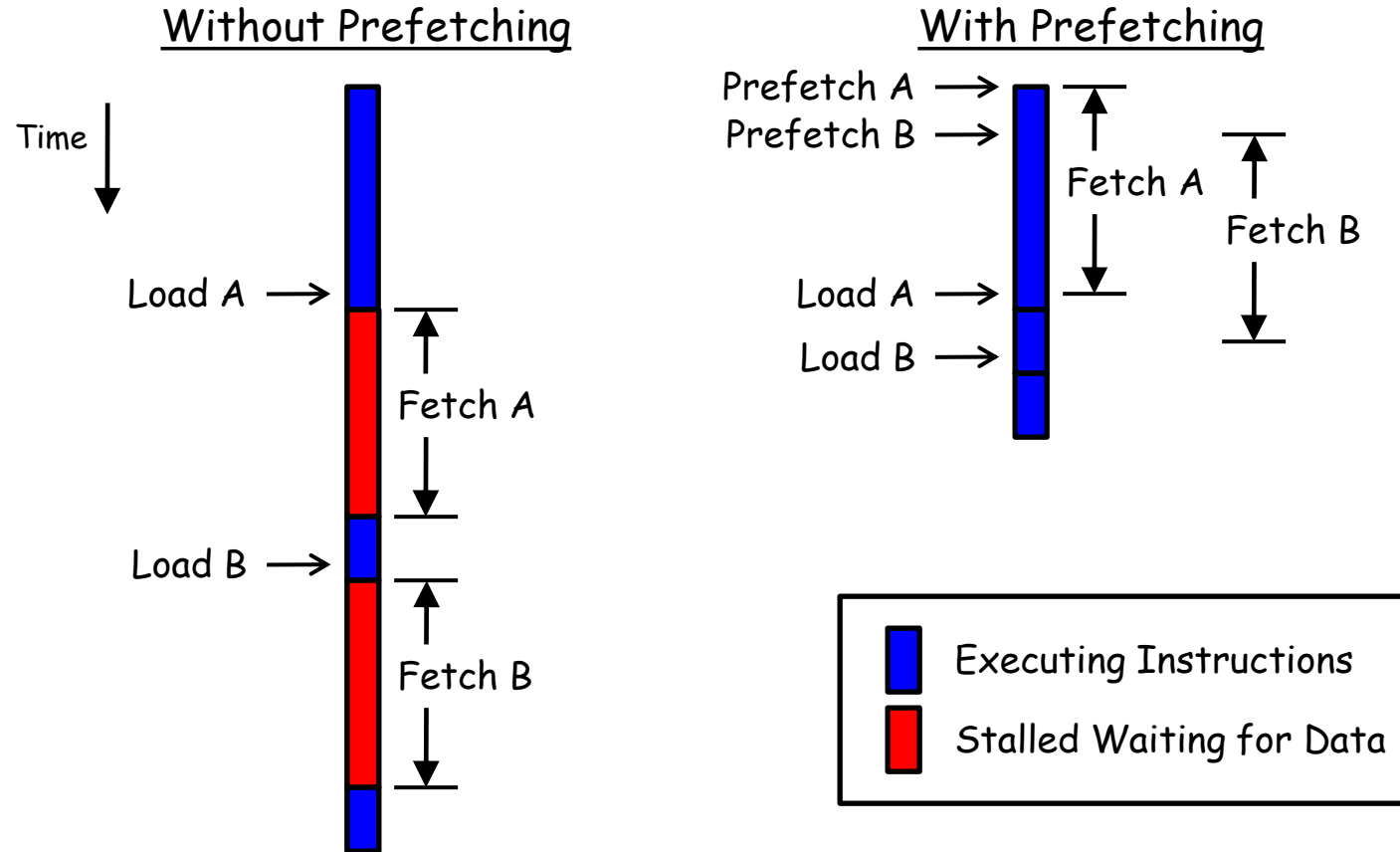
- reorder iterations to improve cache reuse

Tolerate Latency:

— Prefetching

- move data close to the processor before it is needed

Tolerating Latency Through Prefetching



- overlap memory accesses with computation and other accesses

Compiler Algorithm

Analysis: what to prefetch

- Locality Analysis

Scheduling: when/how to issue prefetches

- Loop Splitting
- Software Pipelining

Loop Splitting

- Decompose loops to isolate cache miss instances
 - cheaper than inserting IF statements

Locality Type	Predicate	Loop Transformation
None	True	None
Temporal	$i = 0$	Peel loop i
Spatial	$(i \bmod l) = 0$	Unroll loop i by l

- Apply transformations recursively for nested loops
- Suppress transformations when loops become too large
 - avoid code explosion

Software Pipelining

$$\text{Iterations Ahead} = \left\lceil \frac{l}{s} \right\rceil$$

where l = memory latency, s = shortest path through loop body

Original Loop

```
for (i = 0; i < 100; i++)  
    a[i] = 0;
```

Software Pipelined Loop (5 iterations ahead)

```
for (i = 0; i < 5; i++)          /* Prolog */  
    prefetch(&a[i]);  
  
for (i = 0; i < 95; i++) {      /* Steady State */  
    prefetch(&a[i+5]);  
    a[i] = 0;  
}  
  
for (i = 95; i < 100; i++)      /* Epilog */  
    a[i] = 0;
```